

Optimization of Multivariable Functions

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Introduction

This note is about solving optimization and inequality problems using multivariable calculus. Since multivariable calculus is an advanced topic at undergraduate level, it is less useful in solving MO problems and should not be considered as the first priority in solving inequality problems. The main purpose of this note is to briefly introduce some useful knowledge in this area, and to point out some important issues if one intends to use the relevant results. One should be extremely careful in doing so, because any small gap in a solution of this kind will make the whole argument fail.

In section 1, we provide all necessary terminologies which are needed in the remainder of the note. In particular, one should understand the meanings of open sets, closed sets, bounded sets and compact sets. The extreme value theorem will be used in most problems. Also, the concept of continuity of multivariable functions is important. But since almost all functions that we will come across are continuous, we need not use the rigorous definition.

Section 2 is about solving optimization problems without constraints. We can use partial derivatives to determine all possible points of extrema. This is one possible generalization of one variable differentiation to the multivariable case. The materials in this section are less useful for our purpose, since in many cases we can obtain the same results by using one variable calculus only.

Section 3 is about solving constrained optimization problems using the Lagrange multiplier rule. This is an extremely useful result in multivariable calculus and has many real life applications. It is also useful in proving many inequalities in the Olympiad. We shall see problems with various features that can be solved by the Lagrange multiplier.

1 Euclidean Spaces of Higher Dimensions

This section is about some preliminary concepts of the Euclidean spaces and multi-variable functions.

Definition 1.1. A *vector*(向量) is an n -tuple $(v_1, v_2, \dots, v_n)$ of real numbers. The set $\mathbb{R}^n$ of all such vectors is called the *Euclidean space*(歐幾里得空間/歐氏空間) of dimension n .

When $n = 1$, we identify (x_1) simply by x_1 , and hence $\mathbb{R}^1$ is the same as $\mathbb{R}$. When $n = 2$, the set $\mathbb{R}^2$ is the usual Euclidean plane. We usually use a boldface letter to denote a vector such as $\mathbf{v}$. A vector $\mathbf{v} = (v_1, v_2, \dots, v_n)$ can be represented by an arrow drawing from the origin to the point $(v_1, v_2, \dots, v_n)$. So sometimes we may also identify a vector by its terminal point, and call it a point. The vector $\mathbf{0} = (0, 0, \dots, 0)$ is called the *zero vector*(零向量).

Definition 1.2. The *length*(長度) or *norm*(範數) of $\mathbf{v} = (v_1, v_2, \dots, v_n)$ is defined by

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

In view of Pythagoras' theorem, this means exactly the length of the segment joining the points $\mathbf{v}$ and $\mathbf{0}$. It is also common to use the notation $|\mathbf{v}|$ since the norm is simply the absolute value when $n = 1$. Using this notation, it is not hard to see that the distance between two points $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ is given by

$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}.$$

1.1 Topological Concepts in Euclidean Spaces

In this subsection, we define several topological concepts in $\mathbb{R}^n$. These terminologies will appear in some of the definitions and results in later part of the note, so one should better have a rough understanding on these concepts.

Definition 1.3. For any $\mathbf{a} \in \mathbb{R}^n$ and $r > 0$, the *open ball*(開球) in $\mathbb{R}^n$ with centre $\mathbf{a}$ and radius r is the set

$$B(\mathbf{a}, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| < r\}.$$

The *closed ball*(閉球) in $\mathbb{R}^n$ with centre $\mathbf{a}$ and radius r is the set

$$\overline{B}(\mathbf{a}, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| \leq r\}.$$

This means balls are collection of points which have bounded distances from a fixed centre $\mathbf{a}$, which agrees with the usual meaning of balls in $\mathbb{R}^3$. When $n = 1$, open balls and closed balls are actually open intervals and closed intervals. So these are generalizations of intervals in higher dimensional cases. When $n = 2$, open balls and closed balls are also known as *open discs*(開圓盤) and *closed discs*(閉圓盤) respectively because of their geometric shapes.

Definition 1.4. Let X be a subset of $\mathbb{R}^n$. A point $\mathbf{a} \in X$ is called an *interior point*(內點) of X if there exists $r > 0$ such that $B(\mathbf{a}, r) \subset X$. The *interior*(內部) of X is the set of all interior points of X .

Definition 1.5. Let X be a subset of $\mathbb{R}^n$. A point $\mathbf{a} \in \mathbb{R}^n$ is called a *boundary point*(邊界點) of X if every open ball with centre $\mathbf{a}$ contains some points in X as well as some points not in X . The *boundary*(邊界) of X is the set of all boundary points of X .

These are more rigorous ways to define the concepts of interior and boundary of some sets in $\mathbb{R}^n$. They agree with our intuitive understanding on these terminologies. Note that an interior point of X must belong to X , while a boundary point of X needs not belong to X . It just needs to be sufficiently close to some points in X .

Definition 1.6. A subset X of $\mathbb{R}^n$ is called an *open set*(開集) if every point in X is an interior point of X .

Definition 1.7. A subset X of $\mathbb{R}^n$ is called a *closed set*(閉集) if the complement $\mathbb{R}^n - X$ is an open set.

Open balls and closed balls that we define previously are open sets and closed sets respectively. For the case $n = 1$, it is known that $X \subset \mathbb{R}$ is open if and only if (iff) X is a (finite or infinite) union of disjoint open intervals (possibly of the form (a, ∞) or $(-\infty, b)$).

One intuitive way to understand the meanings of open sets and closed sets is as follows. An open set is a set which does not contain any of its boundary points. A closed set is a set which contains all of its boundary points. For example, the square

$$\{(x, y) : 0 < x < 1, 0 < y < 1\}$$

is open because it does not include any of its sides (which form its boundary). The region

$$\{(x, y) : 2x^2 + 3y^2 \leq 1\}$$

enclosed by an ellipse is closed because it includes all points on the ellipse $2x^2+3y^2=1$. In addition, the ellipse itself is also closed (one can verify that every point on the ellipse is a boundary point of the ellipse). Loosely speaking, if a set of points is defined in a way that the variables satisfy some inequalities or equalities, then a strict inequality sign usually means some boundary points are excluded, while a non-strict inequality sign or an equality sign usually means some boundary points are included. In view of this, the set

$$\{(x, y, z) : 2 \leq x + 2y + 3z < 6\}$$

is neither open nor closed, because it contains some of its boundary points (those satisfying $x + 2y + 3z = 2$) but not all (those satisfying $x + 2y + 3z = 6$ are excluded).

Note that openness and closedness are not mutually inclusive concepts. In $\mathbb{R}^n$, there are two sets, $\emptyset$ and $\mathbb{R}^n$, which are both open and closed.

In general, for any subset X of $\mathbb{R}^n$, we can define open sets and closed sets in X . For example, a subset Y of X is said to be open in X if we can express Y as $U \cap X$ for some open set U in $\mathbb{R}^n$. Interestingly, an open set in X needs not be an open set in $\mathbb{R}^n$. One example is the subset $Y = (0, 1]$ of $X = [-1, 1]$. By definition, Y is open in X because $Y = (0, 2) \cap X$ where $(0, 2)$ is an open interval, but Y is not open in $\mathbb{R}$. In view of this, one should better say that a set is open (or closed) in another set. If we simply say that a set is open, we usually mean it is open in $\mathbb{R}^n$.

Definition 1.8. A subset X of $\mathbb{R}^n$ is called a *bounded set* (有界集) if there exists $M > 0$ such that $\|\mathbf{x}\| \leq M$ for any $\mathbf{x} \in X$.

In other words, every point in a bounded set has a bounded norm, so that it cannot be too far away from the origin. Equivalently, a set is bounded if there exist $M_1, M_2, \dots, M_n$ such that $|x_j| \leq M_j$ for $j = 1, 2, \dots, n$ for every $\mathbf{x} = (x_1, x_2, \dots, x_n)$ inside the set.

Definition 1.9. A subset X of $\mathbb{R}^n$ is called a *compact set* (紧集) if it is closed and bounded.

The topological definition of compact sets is far different from the one given above. However, it can be shown that the definition is the same in $\mathbb{R}^n$.

Finally, the following concept is more related to limits.

Definition 1.10. Let X be a subset of $\mathbb{R}^n$. We say that $\mathbf{a} \in \mathbb{R}^n$ is a *limit point* (極限點) or *accumulation point* (聚點) of X if the following holds. For any $\delta > 0$, there exists $\mathbf{x}$ in the set $X \cap B(\mathbf{a}, \delta)$ which is different from $\mathbf{a}$.

Loosely speaking, $\mathbf{a}$ is a limit point of X if there are many points in X which are sufficiently close to $\mathbf{a}$. Note that $\mathbf{a}$ needs not belong to X . If $\mathbf{a} \notin X$, then $\mathbf{a}$ must belong to the boundary of X . On the other hand, a point $\mathbf{a} \in X$ is said to be an *isolated point*(孤立點) of X if it is not a limit point. Equivalently, this means there exists $\delta > 0$ such that $B(\mathbf{a}, \delta) \cap X = \{\mathbf{a}\}$.

1.2 Multivariable Functions

A function $f(x_1, x_2, \dots, x_n)$ in several real variables can be considered as a function defined on $\mathbb{R}^n$. For our purpose, we are more interested in functions of the form $\mathbf{f} : X \rightarrow \mathbb{R}^m$, where X is a subset of some $\mathbb{R}^n$. Here, we use a boldface letter $\mathbf{f}$ to emphasize that the output is a vector in $\mathbb{R}^m$. As $\mathbf{f}(\mathbf{x})$ has m coordinates for every $\mathbf{x} \in \mathbb{R}^n$, we can write

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$$

for some functions $f_1, f_2, \dots, f_m$ where $f_j : \mathbb{R}^n \rightarrow \mathbb{R}$. The functions $f_1, f_2, \dots, f_m$ are called the *component functions*(分量函數) of $\mathbf{f}$.

For example, consider $\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$\mathbf{f}(x, y, z) = (x^2y, x + y + z).$$

Its component functions are $f_1 : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $f_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by $f_1(x, y, z) = x^2y$ and $f_2(x, y, z) = x + y + z$ respectively. In view of this, to study a function of the form $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, it suffices to study functions of the form $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

When $n = 1$, the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is the usual one variable real-valued function. Analogous to this special case, we can define limits and continuity of a multivariable function.

Definition 1.11. Let $\mathbf{f} : X \rightarrow \mathbb{R}^m$ be a function where $X \subset \mathbb{R}^n$, and let $\mathbf{a}$ be a limit point of X . We say that the *limit*(極限) of $\mathbf{f}$ at $\mathbf{a}$ is $\mathbf{L} \in \mathbb{R}^m$ if the following holds. For any $\varepsilon > 0$, there exists $\delta > 0$ such that if $\mathbf{x} \in X$ and $0 < \|\mathbf{x} - \mathbf{a}\| < \delta$, then $\|\mathbf{f}(\mathbf{x}) - \mathbf{L}\| < \varepsilon$.

In that case, we write

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{L}.$$

As usual, we say that $\mathbf{f}$ tends to $\mathbf{L}$ (which is a vector) when $\mathbf{x}$ tends to $\mathbf{a}$, or write $\mathbf{f}(\mathbf{x}) \rightarrow \mathbf{L}$ when $\mathbf{x} \rightarrow \mathbf{a}$. Loosely speaking, when $\mathbf{x}$ is close to $\mathbf{a}$ but not equal to $\mathbf{a}$, then $\mathbf{f}(\mathbf{x})$ is close to $\mathbf{L}$.

Like the limits of one variable functions, the following properties are satisfied. Suppose $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{L}$ and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{g}(\mathbf{x}) = \mathbf{L}'$.

- The limit $\mathbf{L}$ is uniquely determined.
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{k} = \mathbf{k}$ for any $\mathbf{k} \in \mathbb{R}^m$
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{x} = \mathbf{a}$
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (\mathbf{f} + \mathbf{g})(\mathbf{x}) = \mathbf{L} + \mathbf{L}'$
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (c\mathbf{f})(\mathbf{x}) = c\mathbf{L}$ for any $c \in \mathbb{R}$
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} (fg)(\mathbf{x}) = LL'$ if $f, g: X \rightarrow \mathbb{R}$
- $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \left(\frac{f}{g} \right)(\mathbf{x}) = \frac{L}{L'}$ if $f, g: X \rightarrow \mathbb{R}$ and $L' \neq 0$

More importantly, if $f_1, f_2, \dots, f_m$ are the component functions of the function $\mathbf{f}$, then $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{f}(\mathbf{x}) = (L_1, L_2, \dots, L_m)$ iff $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f_j(\mathbf{x}) = L_j$ for $j = 1, 2, \dots, m$. This again suggests we can focus on real-valued functions.

Definition 1.12. Let $\mathbf{f}: X \rightarrow \mathbb{R}^m$ be a function where $X \subset \mathbb{R}^n$, and let $\mathbf{a} \in X$. We say that $\mathbf{f}$ is *continuous*(連續) at $\mathbf{a}$ if $\mathbf{a}$ is an isolated point, or

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{f}(\mathbf{x}) = \mathbf{f}(\mathbf{a}).$$

Otherwise, we say that $\mathbf{f}$ is *discontinuous*(不連續) at $\mathbf{a}$.

We say that $\mathbf{f}$ is a *continuous function*(連續函數) if it is continuous at every point in X .

Using notations in topology, we can rewrite the definition of continuity (at $\mathbf{a}$) as follows. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\mathbf{f}(X \cap B(\mathbf{a}, \delta)) \subset B(\mathbf{f}(\mathbf{a}), \varepsilon)$. Here, $\mathbf{f}(Y) = \{\mathbf{f}(\mathbf{y}) : \mathbf{y} \in Y\}$ means the image set of Y .

In most cases, we will not come across sets with isolated points. Therefore, the main definition is the limit relation. Most properties of continuous functions follow from those of limits. For example, a function $\mathbf{f}$ is continuous iff all its component functions are continuous. In addition, it can be shown that the composition of two continuous functions is continuous.

Proposition 1.1. Let $\mathbf{f}: X \rightarrow \mathbb{R}^m$ be a continuous function where $X \subset \mathbb{R}^n$, and let Y be a subset of $\mathbb{R}^m$. Define $\mathbf{f}^{-1}(Y) = \{\mathbf{x} \in X : \mathbf{f}(\mathbf{x}) \in Y\}$. Then the following hold.

- If Y is open in $\mathbb{R}^m$, then $\mathbf{f}^{-1}(Y)$ is open in X .
- If Y is closed in $\mathbb{R}^m$, then $\mathbf{f}^{-1}(Y)$ is closed in X .

Proof.

- (a) Suppose Y is open. We have to show that every point in $A = \mathbf{f}^{-1}(Y)$ is an interior point. Pick any $\mathbf{a} \in A$. This means $\mathbf{f}(\mathbf{a}) = \mathbf{y} \in Y$ by definition. Since Y is open, there exists $r > 0$ such that $B(\mathbf{y}, r) \subset Y$. Using the definition of continuity, we can choose $\delta > 0$ such that $\mathbf{f}(X \cap B(\mathbf{a}, \delta)) \subset B(\mathbf{y}, r)$. This implies $\mathbf{f}(X \cap B(\mathbf{a}, \delta)) \subset Y$, and hence $(X \cap B(\mathbf{a}, \delta)) \subset A$. This means $\mathbf{a}$ is an interior point of A as desired.
- (b) Suppose Y is closed. Then $\mathbb{R}^m - Y$ is open by definition. By part (a), the set $\mathbf{f}^{-1}(\mathbb{R}^m - Y)$ is open in X . Note that $\mathbf{f}^{-1}(\mathbb{R}^m - Y) = X - \mathbf{f}^{-1}(Y)$ as $\mathbf{f}^{-1}(\mathbb{R}^m - Y)$ and $\mathbf{f}^{-1}(Y)$ form a partition of X (every point in X is either mapped to $\mathbb{R}^m - Y$ or Y). Again by definition, this means $\mathbf{f}^{-1}(Y)$ is closed in X .

□

This proposition can be used to justify our understanding on open sets and closed sets. For example, to prove that the set

$$X = \{(x, y) : 2x^2 + 3y^2 \leq 1\}$$

is closed in $\mathbb{R}^2$, we define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) = 2x^2 + 3y^2$. Then the set X is exactly the preimage set of the closed interval $(-\infty, 1]$. By [proposition 1.1\(b\)](#), we know that X is closed.

Note that it is not true that the image set of an open/closed set is open/closed. We only have the result for the preimage sets.

Proposition 1.2. Let $\mathbf{f} : X \rightarrow \mathbb{R}^m$ be a continuous function where $X \subset \mathbb{R}^n$, and let C be a compact subset of X . Then $\mathbf{f}(C)$ is a compact set in $\mathbb{R}^m$.

The proof of this result is omitted. Rather than that, we would like to use this to prove the following important result.

Theorem 1.1. ([Extreme value theorem](#)(極値定理)) Let $f : X \rightarrow \mathbb{R}$ be a continuous function where $X \subset \mathbb{R}^n$ is compact. Then f attains its maximum and minimum values.

Proof. By [proposition 1.2](#), we know that $f(X)$ is a compact set in $\mathbb{R}$. In particular, $f(X)$ is bounded. Therefore, it has a smallest upper bound M . (Intuitively, such an M should exist. And indeed this is an axiom of the real number system which cannot be proved.)

If $M \notin f(X)$, then M belongs to the open set $\mathbb{R} - f(X)$ (it is open because $f(X)$ is closed). Thus, there exists ε such that $(M - \varepsilon, M + \varepsilon) \subset (\mathbb{R} - f(X))$ by definition.

But this means no real number in $(M - \varepsilon, M + \varepsilon)$ belongs to $f(X)$, and so $M - \frac{\varepsilon}{2}$ is a smaller upper bound for $f(X)$ than M . This violates the choice of M .

Therefore, we must have $M \in f(X)$. This means $f(\mathbf{a}) = M$ for some $\mathbf{a} \in X$. As M is an upper bound for $f(X)$, we have $f(\mathbf{a}) = M \geq f(\mathbf{x})$ for any $\mathbf{x} \in X$. This proves M is the maximum value of f . Similarly, f attains its minimum value. $\square$

The existence of optimum values is crucial in solving certain inequality problems. Without showing such optimum values exist, sometimes the whole proof fails.

2 Partial Derivatives and Optimization

In this section, we are going to discuss how to use multivariable calculus to find the maximum and minimum values (if exist) of a function $f : X \rightarrow \mathbb{R}$, where X is a subset of $\mathbb{R}^n$. Recall that in one variable calculus, for a function $f : (a, b) \rightarrow \mathbb{R}$ and any $c \in (a, b)$, we define the derivative of f at c by

$$\frac{df}{dx}(c) = f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}.$$

We can use this to determine the monotonicity of f . A first natural generalization of this concept to multivariable functions is given below.

Definition 2.1. Let $f : X \rightarrow \mathbb{R}$ be a function where $X \subset \mathbb{R}^n$, and let $\mathbf{a}$ be an interior point of X . The *partial derivative* (偏導數) of f at $\mathbf{a}$ with respect to (w.r.t.) a variable x_j is defined as

$$\frac{\partial f}{\partial x_j}(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{e}_j) - f(\mathbf{a})}{h},$$

where $\mathbf{e}_j$ is the vector whose j th entry is 1 and all other entries are 0. Another common notation is $f_{x_j}(\mathbf{a})$.

For example, if $f(x, y)$ is a function in two variables, then there are two partial derivatives of f , which are

$$\begin{aligned} f_x(a, b) &= \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}, \\ f_y(a, b) &= \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}. \end{aligned}$$

In short, f_x is the derivative of f w.r.t. the variable x , etc., where all other variables are regarded as constants. As a simple example, let $f(x, y) = y(2x + 3y)^2$. To compute f_x , it suffices to regard y as a constant and obtain

$$\frac{\partial f}{\partial x} = y \frac{\partial}{\partial x} (2x + 3y)^2 = 2y(2x + 3y) \frac{\partial}{\partial x} (2x + 3y) = 4y(2x + 3y)$$

by using the usual differentiation rules such as the chain rule. Similarly, we have

$$\frac{\partial f}{\partial y} = y \frac{\partial}{\partial y} (2x + 3y)^2 + (2x + 3y)^2 \frac{\partial y}{\partial y} = 6y(2x + 3y) + (2x + 3y)^2.$$

Definition 2.2. Let $f : X \rightarrow \mathbb{R}$ be a function where $X \subset \mathbb{R}^n$, and let $\mathbf{a}$ be an interior point of X . Suppose $f_{x_j}(\mathbf{a})$ exists for $j = 1, 2, \dots, n$. The *gradient* (梯度) of f at $\mathbf{a}$ is defined by

$$\nabla f(\mathbf{a}) = (f_{x_1}(\mathbf{a}), f_{x_2}(\mathbf{a}), \dots, f_{x_n}(\mathbf{a})).$$

In other words, the gradient is a vector containing all the partial derivatives at a point. Note that ∇f can be considered as a function (if all partial derivatives exist).

Our goal is to use partial derivatives to determine some points of extrema. First of all, we provide the rigorous definitions of different kinds of extrema.

Definition 2.3. Let $f : X \rightarrow \mathbb{R}$ be a function where $X \subset \mathbb{R}^n$. We say that f has a *local minimum* (局部最小值) at $\mathbf{a} \in X$ if there exists an open set U containing $\mathbf{a}$ such that $f(\mathbf{x}) \geq f(\mathbf{a})$ for any $\mathbf{x} \in (X \cap U)$.

Also, we say that f has a *global minimum* (全局最小值) at $\mathbf{a}$ if $f(\mathbf{x}) \geq f(\mathbf{a})$ for any $\mathbf{x} \in X$.

Similarly, we can define *local maximum* (局部最大值) and *global maximum* (全局最大值). If $\mathbf{a}$ is a point of minimum or maximum, then we say that it is a point of (local or global) *extremum* (極值). For the purpose of solving inequality problems, we are more interested in finding the global extrema. We usually just write minimum or maximum to mean the global ones.

Proposition 2.1. Let $f : X \rightarrow \mathbb{R}$ be a function where $X \subset \mathbb{R}^n$, and let $\mathbf{a}$ be an interior point of X . If f has a local extremum at $\mathbf{a}$ and $\nabla f(\mathbf{a})$ exists, then $\nabla f(\mathbf{a}) = \mathbf{0}$.

Proof. Let $\mathbf{a} = (a_1, a_2, \dots, a_n)$. Define $g(x) = f(x, a_2, a_3, \dots, a_n)$. Since f has a local extremum at $\mathbf{a}$, it is obvious that g has a local extremum at a_1 . Using one variable calculus, we know that $g'(a_1) = 0$. This means exactly $f_{x_1}(\mathbf{a}) = 0$. Similarly, all partial derivatives of f at $\mathbf{a}$ are zero, and hence $\nabla f(\mathbf{a}) = \mathbf{0}$. $\square$

This result gives us a simple way to find out all extremum points. Suppose it is known that f attains a maximum value, and its gradient exists everywhere. Then we just need to solve the equation $\nabla f(\mathbf{x}) = \mathbf{0}$ to find out all possible points of maximum. Among these, the one with the largest image must be the global maximum point.

On the other hand, it is no use to solve the equation $\nabla f(\mathbf{x}) = \mathbf{0}$ if we do not know if f has global extrema or not. Very often, no point is awarded if there is such a gap in a proof using calculus.

Problem 2.1. Let $n \geq 2$, and let $x_1, x_2, \dots, x_n$ be positive real numbers satisfying $x_1 + x_2 + \dots + x_n = 1$. Find the maximum value of $x_1 x_2 \dots x_n$.

Analysis. This is an elementary problem, and can be solved by many simple methods such as the AM-GM inequality. We demonstrate how to use calculus to solve this problem.

First of all, we should not find the gradient of $f(x_1, x_2, \dots, x_n) = x_1 x_2 \dots x_n$ since there is a constraint $x_1 + x_2 + \dots + x_n = 1$, which means **proposition 2.1** cannot be used

directly. A simple way is to replace one of the variables by the other variables so as to remove the constraint.

One great barrier in doing such optimization problems is to show that the maximum value exists. The major result that we have is the extreme value theorem. However, it can only be used if the domain of the function is compact. In this problem, the domain is not compact because of the conditions $x_j > 0$. A useful idea is to extend the domain so that it becomes a compact set.

Solution. The problem is equivalent to finding the maximum value of

$$f(x_1, x_2, \dots, x_{n-1}) = x_1 x_2 \cdots x_{n-1} (1 - x_1 - x_2 - \cdots - x_{n-1})$$

where $x_1, x_2, \dots, x_{n-1} > 0$ and $x_1 + x_2 + \cdots + x_{n-1} < 1$. Let

$$S = \{(x_1, x_2, \dots, x_{n-1}) : x_1, x_2, \dots, x_{n-1} \geq 0, x_1 + x_2 + \cdots + x_{n-1} \leq 1\}.$$

Clearly, this set is closed. It is also bounded because $0 \leq x_j \leq 1$ for each j . Therefore, S is a compact set. Note that f is well-defined on S . As f is continuous, it attains the maximum value on S by the extreme value theorem. Let $\mathbf{a} = (a_1, a_2, \dots, a_{n-1})$ be a point of global maximum.

If $\mathbf{a}$ is an interior point of S , then $a_j > 0$ for each j and $a_1 + a_2 + \cdots + a_{n-1} < 1$. Note that

$$f_{x_1}(x_1, x_2, \dots, x_{n-1}) = x_2 x_3 \cdots x_{n-1} (1 - 2x_1 - x_2 - x_3 - \cdots - x_{n-1}).$$

By [proposition 2.1](#), we must have $f_{x_1}(\mathbf{a}) = 0$. Since $a_j \neq 0$, this implies

$$1 - 2a_1 - a_2 - a_3 - \cdots - a_{n-1} = 0, \quad (1)$$

or $a_1 = 1 - a_1 - a_2 - \cdots - a_{n-1}$. Similarly, we have $a_2 = 1 - a_1 - a_2 - \cdots - a_{n-1} = a_1$. By symmetry, all entries of $\mathbf{a}$ are the same. Thus, (1) becomes $1 - na_1 = 0$. This yields $a_1 = a_2 = \cdots = a_{n-1} = \frac{1}{n}$, and we have

$$f(\mathbf{a}) = \frac{1}{n^n}. \quad (2)$$

If $\mathbf{a}$ is a boundary point of S , then $a_j = 0$ for some j or $a_1 + a_2 + \cdots + a_{n-1} = 1$. In any case, we have $f(\mathbf{a}) = 0$. Comparing with (2), it is clear that 0 cannot be the maximum. Therefore, the maximum value is $\frac{1}{n^n}$. Equality holds when $x_1 = x_2 = \cdots = x_n = \frac{1}{n}$. $\square$

Remarks. The above solution is not at all simple. The main reason is that we spend a lot of effort in showing the maximum value exists (or showing the maximum value is not attained at some boundary point). Indeed, when we use multivariable calculus in practice, we normally do not care too much about this rigour and simply assume the extrema exist. But definitely we cannot do so when solving MO problems. This is one reason why using calculus is not the first priority, even though it is extremely useful in real life application. On the other hand, if we can solve the equation $\nabla f(\mathbf{x}) = \mathbf{0}$, then it still tells us the possible equality cases, even if we cannot show the existence of extrema.

Problem 2.2. Let x, y, z be real numbers such that $0 \leq x, y, z \leq 1$ and they are not all zero. Prove that

$$\frac{1}{1+xy} + \frac{1}{1+yz} + \frac{1}{1+zx} \leq \frac{5}{x+y+z}.$$

Analysis. This time we see how to use multivariable calculus to prove an inequality. Inequality problems can be easily transformed to optimization problems. For instance, we can put everything on the same side and show that the minimum value of the expression is 0. This time we choose to multiply both sides by $x+y+z$ so as to get rid of the missing point $(0,0,0)$.

Solution. Let

$$f(x, y, z) = (x+y+z) \left(\frac{1}{1+xy} + \frac{1}{1+yz} + \frac{1}{1+zx} \right)$$

where $0 \leq x, y, z \leq 1$ (we allow $x = y = z = 0$). Since f is continuous and its domain is compact, it attains the maximum value at some point $\mathbf{a} = (a, b, c)$.

Case 1. $\mathbf{a}$ is an interior point (with $0 < a, b, c < 1$)

It is routine to compute

$$\begin{aligned} f_x(a, b, c) &= \left(\frac{1}{1+ab} + \frac{1}{1+bc} + \frac{1}{1+ca} \right) - (a+b+c) \left(\frac{b}{(1+ab)^2} + \frac{c}{(1+ac)^2} \right), \\ f_y(a, b, c) &= \left(\frac{1}{1+ab} + \frac{1}{1+bc} + \frac{1}{1+ca} \right) - (a+b+c) \left(\frac{a}{(1+ab)^2} + \frac{c}{(1+bc)^2} \right). \end{aligned}$$

Since both are 0 by [proposition 2.1](#), we have $f_x(a, b, c) = f_y(a, b, c)$. Cancelling like terms, we obtain

$$\frac{b}{(1+ab)^2} + \frac{c}{(1+ac)^2} = \frac{a}{(1+ab)^2} + \frac{c}{(1+bc)^2}.$$

Note that $a = b$ must be a factor. Thus, we rewrite this as follows.

$$\begin{aligned} \frac{b}{(1+ab)^2} + \frac{c}{(1+ac)^2} &= \frac{a}{(1+ab)^2} + \frac{c}{(1+bc)^2} \\ \Leftrightarrow \frac{b-a}{(1+ab)^2} &= \frac{c}{(1+bc)^2} - \frac{c}{(1+ac)^2} \\ \Leftrightarrow \frac{b-a}{(1+ab)^2} &= \frac{c^2(a-b)(2+ac+bc)}{(1+ac)^2(1+bc)^2}. \end{aligned}$$

This implies $a = b$ or $\frac{1}{(1+ab)^2} = -\frac{c^2(2+ac+bc)}{(1+ac)^2(1+bc)^2}$. The latter case is rejected since the left-hand side is positive while the right-hand side is negative. By symmetry, we must have $a = b = c$, and hence

$$f_x(a, a, a) = \frac{3}{1+a^2} - \frac{6a^2}{(1+a^2)^2}.$$

As this is 0, we solve

$$\frac{3}{1+a^2} = \frac{6a^2}{(1+a^2)^2}.$$

This means $1+a^2 = 2a^2$, and hence $a = 1$ (rejected). So this case cannot happen.

Case 2. $\mathbf{a}$ is a boundary point with $c = 0$ (or $a = 0$ or $b = 0$)

In this case $(a, b, 0)$ is a point of maximum of

$$f(x, y, 0) = (x + y) \left(\frac{1}{1 + xy} + 2 \right).$$

Again, we have a few subcases to consider.

- If $0 < a, b < 1$, we compute

$$\begin{aligned} f_x(a, b, 0) &= \left(\frac{1}{1 + ab} + 2 \right) - \frac{b(a + b)}{(1 + ab)^2}, \\ f_y(a, b, 0) &= \left(\frac{1}{1 + ab} + 2 \right) - \frac{a(a + b)}{(1 + ab)^2}. \end{aligned}$$

Since both are 0, by comparing these equations, it is obvious that $a = b$. And then we solve

$$f_x(a, a, 0) = \frac{1}{1 + a^2} + 2 - \frac{2a^2}{(1 + a^2)^2} = 0.$$

Clearing denominators, this becomes $2a^4 + 3a^2 + 3 = 0$. There is no solution.

- If $b = 0$ (or $a = 0$), we have $f(a, 0, 0) = 3a$. The maximum value is 3, which is attained at $a = 1$.
- If $b = 1$ (or $a = 1$), we have $f(a, 1, 0) = 2a + 3$. The maximum value is 5, which is attained at $a = 1$.

Case 3. $\mathbf{a}$ is a boundary point with $c = 1$ (or $a = 1$ or $b = 1$)

In this case $(a, b, 1)$ is a point of maximum of

$$f(x, y, 1) = (x + y + 1) \left(\frac{1}{1 + xy} + \frac{1}{1 + x} + \frac{1}{1 + y} \right).$$

We may also assume $a, b > 0$ since the case $a = 0$ (or $b = 0$) has been considered in [case 2](#).

- If $0 < a, b < 1$, we compute

$$f_x(a, b, 1) = \left(\frac{1}{1+ab} + \frac{1}{1+a} + \frac{1}{1+b} \right) - (a+b+1) \left(\frac{b}{(1+ab)^2} + \frac{1}{(1+a)^2} \right),$$

$$f_y(a, b, 1) = \left(\frac{1}{1+ab} + \frac{1}{1+a} + \frac{1}{1+b} \right) - (a+b+1) \left(\frac{a}{(1+ab)^2} + \frac{1}{(1+b)^2} \right).$$

As a corollary of the steps in [case 1](#), we know that $a = b$. Then we solve

$$f_x(a, a, 1) = \frac{1}{1+a^2} + \frac{2}{1+a} - (2a+1) \left(\frac{a}{(1+a^2)^2} + \frac{1}{(1+a)^2} \right) = 0.$$

This means $\frac{1-a-a^2}{(1+a^2)^2} + \frac{1}{(1+a)^2} = 0$. Clearing denominators, this becomes

$$(1-a)(3a^2+3a+2) = 0.$$

There is no solution as $0 < a < 1$.

- If $b = 1$ (or $a = 1$), we have $f(a, 1, 1) = (a+2) \left(\frac{2}{a+1} + \frac{1}{2} \right)$. Again, if $0 < a < 1$, then we solve

$$f_x(a, 1, 1) = \left(\frac{2}{a+1} + \frac{1}{2} \right) - \frac{2(a+2)}{(a+1)^2} = 0.$$

After expansion, this becomes $a^2+2a-3=0$. There is no solution since $0 < a < 1$.

Thus, we must have $a = 1$, which gives $f(1, 1, 1) = \frac{9}{2}$.

Summarizing all these, the maximum value must be $\max \left\{ 5, \frac{9}{2} \right\} = 5$. Equality holds when $(x, y, z) = (1, 1, 0)$ or its permutation. $\square$

In many situations, the domain of the function is $\mathbb{R}^n$ or $(\mathbb{R}^+)^n$. These are unbounded sets, and so we cannot conclude the extremum values exist by using the extreme value theorem. In one variable calculus, there is a result which can replace this. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function such that $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = \infty$. Then f has a minimum value. The reason is that, we can use the extreme value theorem to show that f attains a minimum value m on the domain $[-M, M]$, where M is a large positive constant. When M approaches infinity, $f(x)$ is sufficiently large for any $x \notin [-M, M]$ by the conditions. In particular, $f(x) > m$ if M is large enough (although m depends on M , it can never exceed $f(0)$), so that m is indeed the minimum value of f on $\mathbb{R}$.

When we proceed to higher dimensional cases, we should be careful and see how to generalize the result. For example, consider the function

$$f(x, y) = x^2 - 3xy + y^2$$

defined on $\mathbb{R}^2$. It has the property that $\lim_{x \rightarrow \pm\infty} f(x, y) = \infty$ for any $y \in \mathbb{R}$, and $\lim_{y \rightarrow \pm\infty} f(x, y) = \infty$ for any $x \in \mathbb{R}$. However, it does not have a minimum value because $f(x, x) = -x^2$ can be arbitrarily small. Therefore, merely having infinity limits in some sense is insufficient.

The correct way to generalize the result is to use the idea in the above proof. In other words, we first find a compact set X so that f attains a minimum value m on X . Then we show that $f(x) > m$ for any $x \notin X$. Since the choice of X depends on the question, we do not provide an explicit result here, and simply illustrate the idea by an example.

Problem 2.3. Let x and y be real numbers. Find the minimum value of

$$e^{x^2+y^2} \left(x^2 + y^2 - \frac{8}{3}x + 2 \right).$$

Analysis. The exponential term goes to infinity quickly. Therefore, the expression is large if $x^2 + y^2$ is large. Based on the above idea, we may consider a compact set such that $x^2 + y^2$ is bounded. A simple choice is a closed disc.

Solution. Let M be a sufficiently large positive constant, and let $X = \overline{B}(\mathbf{0}, M)$ be the closed disc with centre $(0, 0)$ and radius M . Let

$$f(x, y) = e^{x^2+y^2} \left(x^2 + y^2 - \frac{8}{3}x + 2 \right).$$

By the extreme value theorem, f attains the minimum value m on X .

Consider any point (x, y) such that $x^2 + y^2 \geq M^2$. We want to show that $f(x, y)$ is large. For example, we have a crude estimation

$$f(x, y) = e^{x^2+y^2} \left(\left(x - \frac{4}{3} \right)^2 + y^2 + \frac{2}{9} \right) \geq \frac{2}{9} e^{M^2} > 2 = f(0, 0).$$

Since $(0, 0) \in X$, this implies $f(x, y) > m$.

Now, we have shown that $f(x, y) \geq m$ no matter (x, y) belongs to X or not. This shows m is the global minimum of f . In addition, the point of global minimum cannot be a boundary point of X , because a boundary point (x, y) must satisfy $x^2 + y^2 = M^2$, which we have shown that $f(x, y) > m$. Therefore, we can apply [proposition 2.1](#) to find the interior extremum point. It is routine to compute

$$\begin{aligned} f_x(x, y) &= e^{x^2+y^2} \left(2x - \frac{8}{3} + 2x \left(x^2 + y^2 - \frac{8}{3}x + 2 \right) \right), \\ f_y(x, y) &= e^{x^2+y^2} \left(2y + 2y \left(x^2 + y^2 - \frac{8}{3}x + 2 \right) \right). \end{aligned}$$

We need to solve $f_x(x, y) = f_y(x, y) = 0$. This is the same as

$$2x - \frac{8}{3} + 2x \left(x^2 + y^2 - \frac{8}{3}x + 2 \right) = 0, \quad (1)$$

$$2y + 2y \left(x^2 + y^2 - \frac{8}{3}x + 2 \right) = 0. \quad (2)$$

From (2), we have $y = 0$ or $x^2 + y^2 - \frac{8}{3}x + 2 = -1$.

- If $y = 0$, equation (1) becomes $2x^3 - \frac{16}{3}x^2 + 6x - \frac{8}{3} = 0$, i.e.

$$\frac{2}{3}(x-1)(3x^2 - 5x + 4) = 0.$$

The only solution is $x = 1$, which gives a solution $(x, y) = (1, 0)$.

- if $x^2 + y^2 - \frac{8}{3}x + 2 = -1$, equation (1) becomes $-\frac{8}{3} = 0$, which has no solution.

Since there is only one solution $(x, y) = (1, 0)$, this must be the point of global minimum. It follows that the minimum value of f is $f(1, 0) = \frac{e}{3}$. $\square$

In many classical inequality problems (in 3 variables), the equality case is $a = b = c$. Since the set of equality is unbounded, we cannot use the trick introduced above to prove the existence of an extremum value. For a homogeneous inequality, a simple trick is to write $b = ax$ and $c = ay$ to reduce the number of variables. Then the set of equality becomes bounded, and we can use the same tricks as before.

Problem 2.4. Let a, b, c be positive real numbers. Prove that

$$(a+b)(b+c)(c+a) \geq 8abc.$$

Analysis. This inequality is of course simple. We mainly want to demonstrate how to show the existence of a minimum value when the set of equality case is unbounded.

Solution. Let $b = ax$ and $c = ay$ where $x, y > 0$. The inequality that we need to prove becomes

$$(1+x)(x+y)(y+1) \geq 8xy.$$

Let

$$f(x, y) = (x+y)(x+1)(y+1) - 8xy.$$

We have to find the minimum value of f .

To show that f has a minimum value, we need to focus on some compact set. We first see what happens if some variables are large. Indeed, we have

$$f(x, y) > (x)(x)(y) - 8xy = xy(x-8) \geq 0$$

if $x \geq 8$. Similarly, $f(x, y) > 0$ if $y \geq 8$. This is good enough.

Consider the compact set $X = \{(x, y) : 0 \leq x, y \leq 8\}$. The function f attains its minimum value on X by the extreme value theorem. Let $\mathbf{a} = (a, b)$ be such a point of minimum. For any $(x, y) \notin X$, we have shown that $f(x, y) > 0 = f(1, 1) \geq f(\mathbf{a})$ since $(1, 1) \in X$. Therefore, $\mathbf{a}$ must be the minimum on $\{(x, y) : x, y \geq 0\}$.

- If $\mathbf{a}$ is an interior point of X , we compute

$$\begin{aligned} f_x(a, b) &= (a+1)(b+1) + (a+b)(b+1) - 8b, \\ f_y(a, b) &= (a+1)(b+1) + (a+b)(a+1) - 8a. \end{aligned}$$

Combining $f_x(a, b) = 0 = f_y(a, b)$, we obtain $(a+b)(a-b) = 8(a-b)$. Therefore, we have $a = b$ or $a+b = 8$.

If $a = b$, then $f_x(a, b) = 0$ becomes $(a+1)^2 + 2a(a+1) - 8a = 0$, i.e. $3a^2 - 4a + 1 = 0$. The solutions are $(a, b) = (1, 1), \left(\frac{1}{3}, \frac{1}{3}\right)$, where $f(1, 1) = 0$ and $f\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{8}{27}$.

If $a+b = 8$, we do not bother to solve $f_x(a, b) = 0$. Indeed, it is obvious that

$$f(a, b) = (a+b)(a+1)(b+1) - 8ab = 8(ab + a + b + 1) - 8ab > 0.$$

- If $\mathbf{a}$ is a boundary point with $a = 0$ (or $b = 0$), then

$$f(0, b) = b(b+1) \geq 0.$$

Equality holds when $b = 0$.

- If $\mathbf{a}$ is a boundary point with $a = 8$ (or $b = 8$), we have shown that $f(a, b) > 0$.

Summarizing these, we see that the minimum value is $\min\left\{0, \frac{8}{27}\right\} = 0$. This proves $f(x, y) \geq 0$ for any x, y as desired. Equality holds when $a = b = c$. $\square$

Finally, let us go through one more contest problem.

Problem 2.5. (APMO 2004 Problem 5) Prove that

$$(a^2 + 2)(b^2 + 2)(c^2 + 2) \geq 9(ab + bc + ca)$$

for all real numbers $a, b, c > 0$.

Analysis. The domain of the function is unbounded. With some trial and error, it is not hard to see that equality holds when $a = b = c = 1$, and probably this is the only equality case. Since the left-hand side of the inequality has degree 2 in each variable, while the right-hand side has degree 1, it is likely that we can restrict to a compact set by removing some large points.

Solution. Let

$$f(a, b, c) = (a^2 + 2)(b^2 + 2)(c^2 + 2) - 9(ab + bc + ca).$$

Let M be a sufficiently large constant. Suppose $\max\{a, b, c\} \geq M$. Without loss of generality (WLOG) assume $a \geq b, c$. We can obtain a lower bound for f crudely. For example, we have

$$(a^2 + 2)(b^2 + 2)(c^2 + 2) > a^2((b^2 + 1) + (c^2 + 1)) \geq Ma(2b + 2c) > 9(ab + bc + ca)$$

because $\max\{ab, ac\} > bc$. This implies $f(a, b, c) > 0$. As $f(1, 1, 1) = 0$, the minimum value of f on the compact set $\{(a, b, c) : 0 \leq a, b, c \leq M\}$ must be the minimum value on $\{(a, b, c) : a, b, c \geq 0\}$. Since the minimum cannot be attained when one of a, b, c is equal to M , it remains to consider the interior points and those boundary points with $a = 0$ (or $b = 0$ or $c = 0$).

If $a = 0$, we have

$$f(0, b, c) = 2(b^2 + 2)(c^2 + 2) - 9bc \geq 2(2\sqrt{2}b)(2\sqrt{2}c) - 9bc = 7bc \geq 0$$

by the AM-GM inequality. There is no equality case.

For interior points, we solve $\nabla f(a, b, c) = \mathbf{0}$, where

$$f_a(a, b, c) = 2a(b^2 + 2)(c^2 + 2) - 9(b + c),$$

$$f_b(a, b, c) = 2b(a^2 + 2)(c^2 + 2) - 9(a + c),$$

$$f_c(a, b, c) = 2c(a^2 + 2)(b^2 + 2) - 9(a + b).$$

- If two of a, b, c are equal, say $b = c$, the equations $f_a(a, b, c) = f_b(a, b, c) = 0$ become

$$2a(b^2 + 2)^2 - 18b = 2b(a^2 + 2)(b^2 + 2) - 9(a + b) = 0.$$

Note that $a = b$ is a solution to the first equality. Thus, we rewrite it as follows.

$$\begin{aligned} & 2a(b^2 + 2)^2 - 18b = 2b(a^2 + 2)(b^2 + 2) - 9(a + b) \\ \Leftrightarrow & 2(b^2 + 2)(ab^2 + 2a - a^2b - 2b) = 9(2b - a - b) \\ \Leftrightarrow & 2(b^2 + 2)(a - b)(2 - ab) = 9(b - a) \\ \Leftrightarrow & (a - b)(-2ab^3 + 4b^2 - 4ab + 17) = 0 \end{aligned}$$

This implies $a = b$ or $a = \frac{4b^2 + 17}{2b(b^2 + 2)}$.

- If $a = b$, then the equation $f_a(a, b, c) = 0$ becomes $2a(a^2 + 2)^2 - 18a = 0$. The only positive solution is $a = b = c = 1$, where $f(1, 1, 1) = 0$.

- If $a = \frac{4b^2 + 17}{2b(b^2 + 2)}$, then the equation $f_a(a, b, c) = 0$ becomes

$$\frac{(4b^2 + 17)(b^2 + 2)}{b} - 18b = 0,$$

i.e. $4b^4 + 7b^2 + 34 = 0$. There is no solution.

- If a, b, c are pairwise distinct, we rewrite $f_a(a, b, c) = f_b(a, b, c)$ as follows.

$$\begin{aligned}
& 2a(b^2 + 2)(c^2 + 2) - 9(b + c) = 2b(a^2 + 2)(c^2 + 2) - 9(a + c) \\
\Leftrightarrow & 2(c^2 + 2)(ab^2 + 2a - a^2b - 2b) = 9(b + c - a - c) \\
\Leftrightarrow & 2(c^2 + 2)(a - b)(2 - ab) = 9(b - a) \\
\Leftrightarrow & (a - b)(-2abc^2 + 4c^2 - 4ab + 17) = 0
\end{aligned}$$

This implies

$$-2abc^2 + 4c^2 - 4ab + 17 = 0. \quad (1)$$

By symmetry, we have $-2a^2bc + 4a^2 - 4bc + 17 = 0$. Note that

$$\begin{aligned}
& -2abc^2 + 4c^2 - 4ab + 17 = -2a^2bc + 4a^2 - 4bc + 17 \\
\Rightarrow & (2a^2bc - 2abc^2) + (4c^2 - 4a^2) + (4bc - 4ab) = 0 \\
\Rightarrow & (a - c)(abc - 2a - 2b - 2c) = 0.
\end{aligned}$$

It follows that

$$abc = 2(a + b + c). \quad (2)$$

Using this, equation (1) can be rewritten as

$$-4c(a + b + c) + 4c^2 - 4ab + 17 = 0,$$

which implies

$$ab + bc + ca = \frac{17}{4}. \quad (3)$$

We now show that (2) and (3) cannot hold simultaneously. Indeed, homogenizing the equations, these yield

$$abc = \frac{8}{17}(a + b + c)(ab + bc + ca).$$

But then

$$\frac{8}{17}(a + b + c)(ab + bc + ca) > \frac{8}{17}(3abc) = \frac{24}{17}abc > abc.$$

This shows there is no solution.

It follows that the minimum value of f is $f(1, 1, 1) = 0$. So the inequality is established. $\square$

3 The Method of Lagrange Multiplier

Very often, we have to find the optimum values of a function under some constraints described by some equations. In those situations, we cannot apply [proposition 2.1](#) to determine the possible points of extrema. In [problem 2.1](#), our solution is to express one variable in terms of the others using the constraint so as to rewrite the question. However, this is not always possible, and may generate some clumsy expression even if we can do so.

In this section, we are going to introduce the method of Lagrange multiplier, which is a well-known result used to solve constrained optimization problems.

Theorem 3.1. ([Lagrange multiplier rule](#)(拉格朗日乘子法)) Let $f, g : X \rightarrow \mathbb{R}$ be functions whose partial derivatives are continuous, where $X \subset \mathbb{R}^n$ is open. Define

$$S = \{\mathbf{x} \in X : g(\mathbf{x}) = c\}$$

where c is a constant. If the restriction $f|_S$ of f on S has a local extremum at $\mathbf{a} \in S$ and $\nabla g(\mathbf{a}) \neq \mathbf{0}$, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a}).$$

The restriction $f|_S$ means the function $f|_S : S \rightarrow \mathbb{R}$ defined by $f|_S(\mathbf{x}) = f(\mathbf{x})$. In other words, we are going to optimize f on the domain S rather than its original domain X (equivalently, we want to optimize $f(\mathbf{x})$ given that $g(\mathbf{x}) = c$). However, it is important for us to mention the set X , because the openness of X is an important condition for the rule to work. We shall see how to choose a suitable X in the problems below.

When solving a constrained optimization problem, we first show that the maximum or minimum value exists (probably using the extreme value theorem again). Then we use the Lagrange multiplier rule to determine all possible points of extrema. This includes all solutions to the equation $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$, as well as all solutions to $\nabla g(\mathbf{x}) = \mathbf{0}$. Do not forget the latter case, or otherwise the solution is incomplete (although this case seldom happens as it is some sort of degenerate cases). The scalar λ in the theorem is the so called Lagrange multiplier.

Problem 3.1. Let a, b, c be real numbers satisfying $a^2 + b^2 + c^2 = 1$. Find the optimum values of

$$ab - b^2 + 3c.$$

Analysis. We first use this example to demonstrate the Lagrange multiplier rule. Comparing with the theorem, we should definitely define $f(a, b, c) = ab - b^2 + 3c$ and $g(a, b, c) = a^2 + b^2 + c^2$. Another major step is to find an open set X such that

$S = \{(a, b, c) \in X : g(a, b, c) = 1\}$ is exactly the domain we are interested in. This is simple since there is no further condition. For example, we can simply take $X = \mathbb{R}^3$.

Solution. Define

$$\begin{aligned} f(a, b, c) &= ab - b^2 + 3c, \\ g(a, b, c) &= a^2 + b^2 + c^2. \end{aligned}$$

They are functions defined on the open set $\mathbb{R}^3$. We are optimizing the function f on the set

$$S = \{(a, b, c) \in \mathbb{R}^3 : g(a, b, c) = 1\}.$$

Since S is compact and f is continuous, both global maximum and global minimum exist by the extreme value theorem. We compute

$$\begin{aligned} \nabla f(a, b, c) &= (b, a - 2b, 3), \\ \nabla g(a, b, c) &= (2a, 2b, 2c). \end{aligned}$$

If $\nabla g(a, b, c) = (0, 0, 0)$, then $a = b = c = 0$. This is rejected since $(0, 0, 0) \notin S$. Therefore, by the Lagrange multiplier rule, the extremum points must be the solutions to $\nabla f(a, b, c) = \lambda \nabla g(a, b, c)$, i.e.

$$b = 2\lambda a, \tag{1}$$

$$a - 2b = 2\lambda b, \tag{2}$$

$$3 = 2\lambda c. \tag{3}$$

From (1) and (2), we get $b^2 = 2\lambda ab = a^2 - 2ab$. This implies

$$b = (-1 \pm \sqrt{2})a. \tag{4}$$

Putting (4) into (1), we have $\lambda = \frac{-1 \pm \sqrt{2}}{2}$ or $a = b = 0$.

- If $\lambda = \frac{-1 \pm \sqrt{2}}{2}$, then $c = 3 \pm 3\sqrt{2}$ by (3). Using the relation $a^2 + b^2 + c^2 = 1$, we have

$$a^2 + (3 \mp 2\sqrt{2})a^2 + 27 \pm 18\sqrt{2} = 1.$$

This gives $a^2 = \frac{-44 \mp 31\sqrt{2}}{2} < 0$, which is impossible.

- If $a = b = 0$, then $c = \pm 1$ by $a^2 + b^2 + c^2 = 1$.

It follows that the points of global extrema must be $(0, 0, 1)$ and/or $(0, 0, -1)$. Since $f(0, 0, 1) = 3$ and $f(0, 0, -1) = -3$, the maximum and minimum values must be 3 and -3 respectively. $\square$

Remarks. When we use the Lagrange multiplier rule, apart from the special case $g(\mathbf{x}) = \mathbf{0}$, we mainly need to solve the system $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$ together with the constraint $g(\mathbf{x}) = c$ (do not forget this constraint!). In total, we have $n + 1$ equations (if $\mathbf{x} \in \mathbb{R}^n$) in $n + 1$ unknowns (n entries of $\mathbf{x}$ together with λ). So most likely we can solve the equations and obtain finitely many solutions.

When solving the equation $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$, we are mainly interested in the solutions of $\mathbf{x}$ but not λ . Therefore, it is common to combine some equations to eliminate the variable λ in the first step (if $\lambda \neq 0$), like what we have done in the above solution. Also, after obtaining $(a, b, c) = (0, 0, \pm 1)$, for example, we do not bother to determine the value of λ . Theoretically, it is possible that λ does not exist (if $(a, b, c) = (0, 0, \pm 1)$ are not solutions and should be rejected). However, our purpose is just to determine all possible points of global extrema in this step, so it does not matter if we include some extra points even if they are not solutions to the system.

Problem 3.2. Let a, b, c be nonnegative real numbers satisfying $a + b + c = 1$. Find the optimum values of

$$(a + 1)b^2 + (b + 1)c^2 + (c + 1)a^2.$$

Analysis. In this problem, the main issue is the extra condition $a, b, c \geq 0$. We cannot choose $X = \mathbb{R}^3$ in the Lagrange multiplier rule, because the set

$$S = \{(a, b, c) \in \mathbb{R}^3 : a + b + c = 1\}.$$

includes those points for which some of a, b, c are negative. One may intend to choose $X = (\mathbb{R}_0^+)^3$, where $\mathbb{R}_0^+$ is the set of all nonnegative real numbers. But then this set is not open in $\mathbb{R}^3$. So we cannot use the Lagrange multiplier rule directly. To solve this problem, we can use some similar ideas as before. We can regard those points with some of a, b, c equal to 0 as boundary points and handle them separately.

Solution. Define

$$\begin{aligned} f(a, b, c) &= (a + 1)b^2 + (b + 1)c^2 + (c + 1)a^2, \\ g(a, b, c) &= a + b + c. \end{aligned}$$

First of all, note that the set $S = \{(a, b, c) : a, b, c \geq 0, g(a, b, c) = 1\}$ is compact. So f must have extremum values on S .

Case 1. $a, b, c > 0$

Define $X_1 = (\mathbb{R}^+)^3$. Then the set

$$S_1 = \{(a, b, c) \in X_1 : g(a, b, c) = 1\}$$

is exactly the subset of S we are considering. Since X_1 is open in $\mathbb{R}^3$, we can use the Lagrange multiplier rule. Thus, we compute

$$\begin{aligned} \nabla f(a, b, c) &= (b^2 + 2a(c + 1), c^2 + 2b(a + 1), a^2 + 2c(b + 1)), \\ \nabla g(a, b, c) &= (1, 1, 1) \neq (0, 0, 0). \end{aligned}$$

Thus, we solve

$$b^2 + 2a(c + 1) = \lambda, \quad (1)$$

$$c^2 + 2b(a + 1) = \lambda, \quad (2)$$

$$a^2 + 2c(b + 1) = \lambda. \quad (3)$$

This time we choose to obtain a symmetric expression from these equations, which is another common trick in solving a system of polynomial equations. Indeed, adding (1), (2) and (3), we obtain

$$(a + b + c)^2 + 2(a + b + c) = 3\lambda.$$

Recall that $a + b + c = 1$. So this yields $\lambda = 1$.

Now, equation (1) can be rewritten as $2a(c + 1) = 1 - b^2 = (1 - b)(1 + b)$. Multiplying this to the other two similar equations, and cancelling out $(1 + a)(1 + b)(1 + c)$ from both sides, we obtain

$$8abc = (1 - a)(1 - b)(1 - c).$$

Using $a + b + c = 1$, this can be simplified as $9abc = ab + bc + ca$. Equivalently, we have

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 9.$$

By the Cauchy-Schwarz inequality, we have

$$(1)(9) = (a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9.$$

Since the equality holds, we must have $a = b = c = \frac{1}{3}$.

Case 2. $a = 0$ and $b, c \neq 0$ (or other similar cases)

We need to optimize

$$f(b, c) = b^2 + (b + 1)c^2$$

under the constraint $g(b, c) = b + c = 1$ (we use the same symbols for convenience). As above, we can define the open set $X_2 = (\mathbb{R}^+)^2$ in $\mathbb{R}^2$, and work on the set

$$S_2 = \{(b, c) \in X_2 : b + c = 1\}.$$

Again, we compute $\nabla f(b, c) = (2b + c^2, 2c(b + 1))$ and $\nabla g(b, c) = (1, 1) \neq (0, 0)$. Then we solve

$$\begin{aligned} 2b + c^2 &= \mu, \\ 2c(b + 1) &= \mu. \end{aligned}$$

Combining these equations, we obtain $2b + c^2 = 2c(b + 1)$. Putting $b = 1 - c$ into this equation, we get $2(1 - c) + c^2 = 2c(2 - c)$, i.e. $3c^2 - 6c + 2 = 0$. The solution is $c = 1 - \frac{1}{\sqrt{3}}$

as $0 < c < 1$, and hence $b = \frac{1}{\sqrt{3}}$.

Case 3. $a = b = 0$ (or other similar cases)

Clearly, we must have $(a, b, c) = (0, 0, 1)$.

To sum up, the extremum points can only be $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$, $\left(0, \frac{1}{\sqrt{3}}, 1 - \frac{1}{\sqrt{3}}\right)$ or $(0, 0, 1)$ (or their cyclic permutations). By comparing the values

$$f\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) = \frac{4}{9}, \quad f\left(0, \frac{1}{\sqrt{3}}, 1 - \frac{1}{\sqrt{3}}\right) = 1 - \frac{2}{3\sqrt{3}}, \quad f(0, 0, 1) = 1,$$

we see that the maximum is 1, while the minimum is $\frac{4}{9}$. $\square$

Remarks. The steps of solving the system of equations in **case 1** seem to be artificial. However, when solving polynomial equations which are symmetric or cyclic in the variables, we usually take the sum or product of all equations to retain the symmetry. We may regard the elementary symmetric polynomials, like $a + b + c$, $ab + bc + ca$ and abc , as variables, and solve for their values. Another common trick is to take the difference of some pairs of equations to factor out some linear terms such as $a - b$.

Problem 3.3. (CWMO 2004 Problem 8) Prove that for any positive real numbers a, b, c , we have

$$1 < \frac{a}{\sqrt{a^2 + b^2}} + \frac{b}{\sqrt{b^2 + c^2}} + \frac{c}{\sqrt{c^2 + a^2}} \leq \frac{3\sqrt{2}}{2}.$$

Analysis. The inequality on the left will be left as an exercise since it is not related to the Lagrange multiplier rule. Note that we would not think of the rule because the left inequality is a strict inequality. This probably means the minimum does not exist or we are not able to determine its exact value. (A hint to prove this inequality is to use **proposition 3.1(a)** of the note **Inequalities 5.0**).

For the inequality on the right, we intend to use the Lagrange multiplier rule. However, to do so, we must have a condition like $a + b + c = 1$. Although such a condition is not given in this problem, we can still impose a new condition by substitution (or sometimes by normalization). The substitutions $x = \frac{b^2}{a^2}$, $y = \frac{c^2}{b^2}$, $z = \frac{a^2}{c^2}$ seem to serve this purpose as we have a simple constraint $xyz = 1$, and all terms in the expression are simplified.

Solution. We only prove

$$\frac{a}{\sqrt{a^2 + b^2}} + \frac{b}{\sqrt{b^2 + c^2}} + \frac{c}{\sqrt{c^2 + a^2}} \leq \frac{3\sqrt{2}}{2}$$

in this solution. Let $x = \frac{b^2}{a^2}$, $y = \frac{c^2}{b^2}$ and $z = \frac{a^2}{c^2}$. Then $g(x, y, z) = xyz = 1$ and we need to find the maximum value of

$$f(x, y, z) = \frac{1}{\sqrt{1+x}} + \frac{1}{\sqrt{1+y}} + \frac{1}{\sqrt{1+z}}.$$

Let $X = (\mathbb{R}^+)^3$ be an open set. Then we are maximizing f on the set

$$S = \{(x, y, z) \in X : g(x, y, z) = 1\}.$$

This can be done using the Lagrange multiplier rule (assuming the maximum value exists). We compute

$$\begin{aligned} \nabla f(x, y, z) &= \left(-\frac{1}{2(1+x)^{\frac{3}{2}}}, -\frac{1}{2(1+y)^{\frac{3}{2}}}, -\frac{1}{2(1+z)^{\frac{3}{2}}} \right), \\ \nabla g(x, y, z) &= (yz, zx, xy) \neq (0, 0, 0). \end{aligned}$$

Thus, we solve

$$-\frac{1}{2(1+x)^{\frac{3}{2}}} = \lambda yz, \tag{1}$$

$$-\frac{1}{2(1+y)^{\frac{3}{2}}} = \lambda zx, \tag{2}$$

$$-\frac{1}{2(1+z)^{\frac{3}{2}}} = \lambda xy.$$

Eliminating λ by combining (1) and (2), we get $\frac{x^2}{(1+x)^3} = \frac{y^2}{(1+y)^3}$. Now,

$$\begin{aligned} &\frac{x^2}{(1+x)^3} = \frac{y^2}{(1+y)^3} \\ \Rightarrow &x^2(y^3 + 3y^2 + 3y + 1) = y^2(x^3 + 3x^2 + 3x + 1) \\ \Rightarrow &(y-x)(x^2y^2 - 3xy - x - y) = 0. \end{aligned} \tag{3}$$

If $x = y = z$, then $x = y = z = 1$. If x, y, z are not all equal, WLOG we may assume $x \neq y$ and $x \neq z$. Therefore, equation (3) implies

$$x^2y^2 - 3xy - x - y = 0. \tag{4}$$

By symmetry, we also have

$$x^2z^2 - 3xz - x - z = 0. \tag{5}$$

Taking the difference of (4) and (5), we obtain

$$x^2(y^2 - z^2) - 3x(y - z) - (y - z) = 0,$$

which yields

$$(y - z)(x^2y + x^2z - 3x - 1) = 0.$$

- If we have $y \neq z$ in addition, then $x^2y + x^2z - 3x - 1 = 0$. By symmetry, we have $y^2x + y^2z - 3y - 1 = 0$. Again, we take their difference to get

$$xy(x - y) + z(x^2 - y^2) - 3(x - y) = 0.$$

This implies $(x - y)(xy + yz + zx - 3) = 0$, and hence $xy + yz + zx = 3$. By the AM-GM inequality, we have

$$3 = xy + yz + zx \geq 3\sqrt[3]{(xyz)^2} = 3.$$

Since the equality holds, we must have $x = y = z$, which is a contradiction.

- If $y = z$, then $x = \frac{1}{y^2}$ from $xyz = 1$. Now, equation (4) becomes $\frac{1}{y^2} - \frac{3}{y} - \frac{1}{y^2} - y = 0$. This means $y^2 = -3$, which has no real solution.

Thus, we have seen that $(x, y, z) = (1, 1, 1)$ is the only possible point of global maximum, where $f(1, 1, 1) = \frac{3\sqrt{2}}{2}$ agrees with the final result. Therefore, it remains to prove that the maximum value does exist.

Note that S is not compact as it is not bounded. Using the standard idea as before, it suffices to prove $f(x, y, z) \leq \frac{3\sqrt{2}}{2}$ whenever $\max\{x, y, z\}$ is sufficiently large. WLOG assume $\max\{x, y, z\} = x$. Note that

$$f(x, y, z) = \frac{1}{\sqrt{1+x}} + \frac{1}{\sqrt{1+y}} + \frac{1}{\sqrt{1+z}} < \frac{1}{\sqrt{1+x}} + 2$$

as $y, z > 0$. Since

$$\lim_{x \rightarrow \infty} \left(\frac{1}{\sqrt{1+x}} + 2 \right) = 0 + 2 = 2 < \frac{3\sqrt{2}}{2},$$

the inequality $f(x, y, z) \leq \frac{3\sqrt{2}}{2}$ must hold for sufficiently large x . This completes the proof. $\square$

Problem 3.4. (CGMO 2011 Problem 3) The positive reals a, b, c, d satisfy $abcd = 1$. Prove that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{9}{a+b+c+d} \geq \frac{25}{4}.$$

Analysis. This is another similar problem where the domain is unbounded. The ideas are mostly the same as the previous problems.

Solution. Define

$$f(a, b, c, d) = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{9}{a+b+c+d}$$

and $g(a, b, c, d) = abcd$. Note that f is large if one of a, b, c, d is small. And this happens when one of a, b, c, d is large because of the condition $abcd = 1$. So it is not hard to handle the large points.

If $a \geq M^3$ for some large constant M , then one of b, c, d must be bounded above by $\frac{1}{M}$ since $bcd = \frac{1}{a} \leq \frac{1}{M^3}$. WLOG assume $d \leq \frac{1}{M}$. Then we have

$$f(a, b, c, d) > \frac{1}{d} \geq M.$$

By taking $M > \frac{25}{4}$, the desired inequality holds. So it remains to find the minimum value of f on the compact set

$$\{(a, b, c, d) : 0 < a, b, c, d \leq M^3, g(a, b, c, d) = 1\}.$$

(This set is compact even though there is a strict inequality > 0 . This is because a, b, c, d must be nonzero from $abcd = 1$.) By the extreme value theorem, the minimum value must exist. Also, it cannot be attained at a boundary point with $\max\{a, b, c, d\} = M^3$ from above. So it remains to determine all possible points of minimum of f on the domain

$$S = \{(a, b, c, d) : 0 < a, b, c, d < M^3, g(a, b, c, d) = 1\}.$$

This set can be rewritten as $\{(a, b, c, d) \in X : g(a, b, c, d) = 1\}$ where $X = (0, M^3)^4$ is open in $\mathbb{R}^4$. Therefore, we can use the Lagrange multiplier rule.

We compute $f_a(a, b, c, d) = -\frac{1}{a^2} - \frac{9}{(a+b+c+d)^2}$ (similarly for f_b, f_c, f_d) and

$$\nabla g(a, b, c, d) = (bcd, cda, dab, abc) \neq (0, 0, 0, 0).$$

We solve the system

$$-\frac{1}{a^2} - \frac{9}{(a+b+c+d)^2} = \lambda bcd, \tag{1}$$

$$-\frac{1}{b^2} - \frac{9}{(a+b+c+d)^2} = \lambda cda, \tag{2}$$

$$-\frac{1}{c^2} - \frac{9}{(a+b+c+d)^2} = \lambda dab,$$

$$-\frac{1}{d^2} - \frac{9}{(a+b+c+d)^2} = \lambda abc.$$

Since a times (1) is equal to b times (2), we have

$$\frac{1}{a} + \frac{9a}{(a+b+c+d)^2} = \frac{1}{b} + \frac{9b}{(a+b+c+d)^2}.$$

By symmetry, $a - b$ is a factor. So this can be rewritten as

$$(a - b) \left(\frac{9}{(a+b+c+d)^2} - \frac{1}{ab} \right) = 0. \tag{3}$$

If $a = b = c = d$, then $a = b = c = d = 1$, which is one of the solutions. Otherwise, we may assume $a \neq b$. Then (3) becomes

$$9ab = (a + b + c + d)^2. \quad (4)$$

If in addition $c \neq a$, then similarly $9ac = (a + b + c + d)^2$. Combining these, we obtain $b = c$. In other words, we must have $c = a$ or $c = b$. Similarly, we have $d = a$ or $d = b$. This gives two possibilities up to symmetry.

- If $a = c$ and $b = d$, then $ab = 1$ from $abcd = 1$, and (4) becomes $9 = 4(a + b)^2$. This gives $a + b = \frac{3}{2}$. Hence, a, b are solutions to $x^2 - \frac{3}{2}x + 1 = 0$, which has no real solution.
- If $a = c = d$, then $b = \frac{1}{a^3}$ from $abcd = 1$. Equation (4) becomes $\frac{9}{a^2} = \left(3a + \frac{1}{a^3}\right)^2$. This gives $3a^4 - 3a^2 + 1 = 0$. Again, it has no solution.

Therefore, the point of minimum can only be $(a, b, c, d) = (1, 1, 1, 1)$. This gives $f(a, b, c, d) \geq f(1, 1, 1, 1) = \frac{25}{4}$ as desired. $\square$

Problem 3.5. (IMO Shortlist 2006 A6/Problem 3) Determine the least real number M such that the inequality

$$|ab(a^2 - b^2) + bc(b^2 - c^2) + ca(c^2 - a^2)| \leq M(a^2 + b^2 + c^2)^2$$

holds for all real numbers a, b and c .

Analysis. Although there is no constraint in this problem, we can impose a constraint by normalization. The simplest way is to give a constant value to $a^2 + b^2 + c^2$, since this gives us a compact domain while the method of Lagrange multiplier can be applied directly. In addition, the inequality is simplified.

Solution. It suffices to consider the case when a, b, c are not all zero. Since the inequality is homogeneous, WLOG assume $g(a, b, c) = a^2 + b^2 + c^2 = 1$. We have to find the extremum values of

$$f(a, b, c) = ab(a^2 - b^2) + bc(b^2 - c^2) + ca(c^2 - a^2).$$

Since the set $S = \{(a, b, c) \in \mathbb{R}^3 : g(a, b, c) = 1\}$ is compact, the extrema exist by the extreme value theorem. We compute

$$\begin{aligned} \nabla f(a, b, c) &= (3a^2b - b^3 + c^3 - 3a^2c, 3b^2c - c^3 + a^3 - 3b^2a, 3c^2a - a^3 + b^3 - 3c^2b), \\ \nabla g(a, b, c) &= (2a, 2b, 2c). \end{aligned}$$

If $\nabla g(a, b, c) = (0, 0, 0)$, then $a = b = c = 0$, which contradicts $a^2 + b^2 + c^2 = 1$. Therefore, we only need to solve

$$3a^2b - b^3 + c^3 - 3a^2c = 2\lambda a, \quad (1)$$

$$3b^2c - c^3 + a^3 - 3b^2a = 2\lambda b, \quad (2)$$

$$3c^2a - a^3 + b^3 - 3c^2b = 2\lambda c.$$

Combining (1) and (2), we obtain

$$3a^2b^2 - b^4 + bc^3 - 3a^2bc = 2\lambda ab = 3ab^2c - ac^3 + a^4 - 3a^2b^2.$$

This means

$$a^4 - 6a^2b^2 + b^4 - ac^3 - bc^3 + 3a^2bc + 3ab^2c = 0. \quad (3)$$

Similarly, we have

$$b^4 - 6b^2c^2 + c^4 - ba^3 - ca^3 + 3b^2ca + 3bc^2a = 0. \quad (4)$$

Equating (3) and (4), and taking out the common factor $a - c$, we get

$$(a - c)(a^3 + 2a^2c + 2ac^2 + c^3 + a^2b + 4abc + bc^2 - 6ab^2 - 6b^2c) = 0.$$

Note that if any two of a, b, c are equal, then $f(a, b, c) = 0$, which is obviously neither a maximum nor a minimum. Therefore, we can remove the factor $a - c$. Due to symmetry, we obtain 3 similar equations as follows:

$$a^3 + 2a^2c + 2ac^2 + c^3 + a^2b + 4abc + bc^2 - 6ab^2 - 6b^2c = 0, \quad (5)$$

$$b^3 + 2b^2a + 2ba^2 + a^3 + b^2c + 4bca + ca^2 - 6bc^2 - 6c^2a = 0, \quad (6)$$

$$c^3 + 2c^2b + 2cb^2 + b^3 + c^2a + 4cab + ab^2 - 6ca^2 - 6a^2b = 0. \quad (7)$$

Once again, equating (5) and (6), and taking out the common factor $b - c$, we get

$$(b - c)(b^2 + 8bc + c^2 + 8ab + 8ac + a^2) = 0.$$

Hence, $a^2 + b^2 + c^2 + 8(ab + bc + ca) = 0$. As $a^2 + b^2 + c^2 = 1$, this yields

$$ab + bc + ca = -\frac{1}{8}.$$

It follows that

$$a + b + c = \pm \sqrt{(a^2 + b^2 + c^2) + 2(ab + bc + ca)} = \pm \frac{\sqrt{3}}{2}.$$

Recall that it is always useful to consider the elementary symmetric polynomials. We have found the values of $a + b + c$ and $ab + bc + ca$, so it remains to find the value of abc . For example, one way is to sum up (5), (6) and (7) to get a symmetric equation. This gives

$$2(a^3 + b^3 + c^3) - 3(a^2b + a^2c + b^2a + b^2c + c^2a + c^2b) + 12abc = 0.$$

Note that

$$\begin{aligned} & 2(a^3 + b^3 + c^3) - 3(a^2b + a^2c + b^2a + b^2c + c^2a + c^2b) + 12abc = 0 \\ \Rightarrow & 2(a + b + c)^3 - 9(a + b + c)(ab + bc + ca) + 27abc = 0 \\ \Rightarrow & \pm \frac{3\sqrt{3}}{4} \pm \frac{9\sqrt{3}}{16} + 27abc = 0 \\ \Rightarrow & abc = \mp \frac{7\sqrt{3}}{144}. \end{aligned}$$

Thus, by Vieta's theorem, a, b, c are the roots to the cubic equation

$$x^3 \mp \frac{\sqrt{3}}{2}x^2 - \frac{1}{8}x \pm \frac{7\sqrt{3}}{144} = 0.$$

Putting $x = \sqrt{3}y$, this becomes

$$3y^3 \mp \frac{3}{2}y^2 - \frac{1}{8}y \pm \frac{7}{144} = 0.$$

One may apply further substitutions to simplify the equation or use the rational root theorem to test all possible rational roots. In any case, it is not hard to deduce a root $y = \pm \frac{1}{6}$. Thus, the equation becomes

$$\left(y \mp \frac{1}{6}\right)\left(3y^2 \mp y - \frac{7}{24}\right) = 0.$$

The solutions are $y = \pm \frac{1}{6}, \frac{\pm 2 + 3\sqrt{2}}{12}, \frac{\pm 2 - 3\sqrt{2}}{12}$. Therefore, the extremum values must be attained when $\{a, b, c\} = \left\{\pm \frac{\sqrt{3}}{6}, \frac{\pm 2\sqrt{3} + 3\sqrt{6}}{12}, \frac{\pm 2\sqrt{3} - 3\sqrt{6}}{12}\right\}$.

In order to reduce the amount of computations, we make the following observation. Since $f(a, b, c) = 0$ whenever two of a, b, c are equal, $(a - b)(b - c)(c - a)$ must be a factor of f . Also, f is degree 4 and is cyclic. So the remaining factor must be $a + b + c$ up to a constant factor. Therefore, it is easy to obtain

$$f(a, b, c) = -(a - b)(b - c)(c - a)(a + b + c).$$

Clearly, interchanging any two variables is the same as multiplying -1 to f . So it suffices to consider $(a, b, c) = \left(\pm \frac{\sqrt{3}}{6}, \frac{\pm 2\sqrt{3} + 3\sqrt{6}}{12}, \frac{\pm 2\sqrt{3} - 3\sqrt{6}}{12}\right)$ and compute

$$\begin{aligned} f(a, b, c) &= -\left(\pm \frac{\sqrt{3}}{6} - \frac{\pm 2\sqrt{3} + 3\sqrt{6}}{12}\right)\left(\frac{\pm 2\sqrt{3} + 3\sqrt{6}}{12} - \frac{\pm 2\sqrt{3} - 3\sqrt{6}}{12}\right) \\ &\quad \times \left(\frac{\pm 2\sqrt{3} - 3\sqrt{6}}{12} \mp \frac{\sqrt{3}}{6}\right)\left(\pm \frac{\sqrt{3}}{6} + \frac{\pm 2\sqrt{3} + 3\sqrt{6}}{12} + \frac{\pm 2\sqrt{3} - 3\sqrt{6}}{12}\right) \\ &= \mp \frac{\sqrt{6}}{4} \cdot \frac{\sqrt{6}}{2} \cdot \frac{\sqrt{6}}{4} \cdot \frac{\sqrt{3}}{2} \\ &= \mp \frac{9\sqrt{2}}{32}. \end{aligned}$$

It follows that the extremum values are $\pm \frac{9\sqrt{2}}{32}$. This proves $|f(a, b, c)| \leq \frac{9\sqrt{2}}{32}$ where the equality case can be attained. Thus, the minimum M is $\frac{9\sqrt{2}}{32}$. $\square$

Problem 3.6. (IMO Shortlist 2003 A4/Problem 5) Given $n > 2$ and reals $x_1 \leq x_2 \leq \dots \leq x_n$, show that

$$\left(\sum_{i,j=1}^n |x_i - x_j| \right)^2 \leq \frac{2}{3}(n^2 - 1) \sum_{i,j=1}^n (x_i - x_j)^2.$$

Show that we have equality iff the sequence is an arithmetic progression.

Analysis. If we consider the quotient of the two sides, then the gradient of the resultant function is clumsy. Therefore, a simple way is to assign a constant value to the right-hand side. Again, this can be done because the inequality is homogeneous. In that case, we may further ignore the square on the left-hand side and optimize a simple function. The equality case can be easily found using the Lagrange multiplier rule. The main difficulty in this problem is to show the existence of the optimum values, because the set of equality is unbounded.

One important feature of the given expressions is that their values are unchanged if we shift all variables by the same constant. We can use this feature so as to focus on a bounded region.

Solution. It suffices to consider the case when the variables are not all equal, or otherwise both sides are zero. By normalization, we may assume

$$g(x_1, x_2, \dots, x_n) = \sum_{i,j=1}^n (x_i - x_j)^2 = 1.$$

We need to maximize

$$f(x_1, x_2, \dots, x_n) = \sum_{i,j=1}^n |x_i - x_j| = 2 \sum_{1 \leq i < j \leq n} (x_j - x_i) = 2 \sum_{i=1}^n (2i - n - 1)x_i.$$

In view of the given expressions, it is obvious that both f and g are unchanged if we add every variable by the same constant. Therefore, by choosing a suitable constant, we may assume $m = \min \{x_1, x_2, \dots, x_n\} \leq 0$ and $M = \max \{x_1, x_2, \dots, x_n\} \geq 0$. In that case, from

$$(M - m)^2 \leq \sum_{i,j=1}^n (x_i - x_j)^2 = 1,$$

we clearly have $-1 \leq m \leq 0 \leq M \leq 1$. In other words, the image of every point in

$$S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : g(x_1, x_2, \dots, x_n) = 1\}$$

under f is equal to the image of some point in

$$S_1 = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : -1 \leq x_1, x_2, \dots, x_n \leq 1, g(x_1, x_2, \dots, x_n) = 1\}.$$

As f attains the maximum value on S_1 (because S_1 is compact), it also attains the maximum value on S .

Now, we can apply the Lagrange multiplier rule to maximize f on the domain S . The i th entry of $\nabla f(x_1, x_2, \dots, x_n)$ is $2(2i - n - 1)$, while the i th entry of $\nabla g(x_1, x_2, \dots, x_n)$ is

$$4 \sum_{j=1}^n (x_i - x_j) = 4nx_i - 4 \sum_{j=1}^n x_j.$$

If $\nabla g(x_1, x_2, \dots, x_n) = (0, 0, \dots, 0)$, then $nx_i = x_1 + x_2 + \dots + x_n$ for any i . This implies $x_1 = x_2 = \dots = x_n$, which contradicts with $g(x_1, x_2, \dots, x_n) = 1$. Therefore, we only need to solve the equation $\nabla f = \lambda \nabla g$. This means

$$2(2i - n - 1) = 4\lambda \left(nx_i - \sum_{j=1}^n x_j \right)$$

for $i = 1, 2, \dots, n$. Taking the difference of the i th equation and the j th equation, we can remove the summation. This gives

$$i - j = \lambda n(x_i - x_j)$$

for any $i \neq j$. Obviously, $\lambda \neq 0$. Hence, $x_n - x_{n-1} = x_{n-1} - x_{n-2} = \dots = x_2 - x_1$. This means exactly that $x_1, x_2, \dots, x_n$ form an arithmetic progression. In view of the question statement, all these should be points of maximum. So we just need to show that equality holds if $x_1, x_2, \dots, x_n$ is an arithmetic progression.

Indeed, if $x_i = a + (i - 1)d$ for any $i = 1, 2, \dots, n$ where a and d are fixed, then

$$\begin{aligned} \left(\sum_{i,j=1}^n |x_i - x_j| \right)^2 &= \left(\sum_{i,j=1}^n |i - j| d \right)^2 \\ &= \left(2 \sum_{i=1}^n (2i - n - 1) id \right)^2 \\ &= 4d^2 \left(\frac{n(n+1)(2n+1)}{3} - \frac{n(n+1)^2}{2} \right)^2 \\ &= \frac{d^2(n-1)^2 n^2 (n+1)^2}{9} \end{aligned}$$

and

$$\begin{aligned} \frac{2}{3}(n^2 - 1) \sum_{i,j=1}^n (x_i - x_j)^2 &= \frac{2}{3}(n^2 - 1) \sum_{i,j=1}^n (i - j)^2 d^2 \\ &= \frac{4}{3}(n^2 - 1) d^2 \sum_{k=1}^{n-1} (n - k) k^2 \\ &= \frac{4}{3}(n^2 - 1) d^2 \left(\frac{(n-1)n^2(2n-1)}{6} - \frac{(n-1)^2 n^2}{4} \right) \\ &= \frac{d^2(n-1)^2 n^2 (n+1)^2}{9}. \end{aligned}$$

The two sides are equal, so the inequality is proved. Equality holds iff $x_1, x_2, \dots, x_n$ form an arithmetic progression from above. $\square$

Problem 3.7. (IMO Shortlist 2008 A2/Problem 2)

(a) Prove that

$$\frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2} + \frac{z^2}{(z-1)^2} \geq 1$$

for all real numbers x, y, z , each different from 1, and satisfying $xyz = 1$.

(b) Prove that equality holds above for infinitely many triples of rational numbers x, y, z , each different from 1, and satisfying $xyz = 1$.

Analysis. We go through one more example where the equality cases can be easily deduced using the method of Lagrange multiplier, but it is hard to tell whether the minimum exists or not. The set of equality is unbounded (which can be seen after solving the equation), and there is not special feature of this set unlike the previous problem.

The key idea in the following solution is to keep one of the variables as a constant, say z , and we find the minimum value of the expression for every such z . This is essentially the idea of the local refinement method.

Solution. Suppose we try to apply the Lagrange multiplier rule. We will be solving the following system:

$$-\frac{2x}{(x-1)^3} = \lambda yz, \tag{1}$$

$$-\frac{2y}{(y-1)^3} = \lambda zx, \tag{2}$$

$$-\frac{2z}{(z-1)^3} = \lambda xy.$$

Combining (1) and (2), we get $\frac{2x^2}{(x-1)^3} = -\lambda xyz = \frac{2y^2}{(y-1)^3}$. Note that

$$\begin{aligned} & \frac{2x^2}{(x-1)^3} = \frac{2y^2}{(y-1)^3} \\ \Rightarrow & x^2(y^3 - 3y^2 + 3y - 1) = y^2(x^3 - 3x^2 + 3x - 1) \\ \Rightarrow & (y-x)(x^2y^2 - 3xy + x + y) = 0. \end{aligned}$$

Clearly, the case $x = y = z$ cannot happen (otherwise $x = y = z = 1$). So we may assume $x \neq y$, and deduce $x^2y^2 - 3xy + x + y = 0$. It turns out that equality holds whenever x and y satisfy this relation (given that $x, y, z \neq 1$).

As suggested, we should modify the above argument in order to show the existence of the minimum value. WLOG assume $|x|, |y| \leq |z|$. If $z \geq \frac{1}{2}$, then $\frac{z^2}{(z-1)^2} \geq 1$ (since this is equivalent to $2z \geq 1$), and the inequality must hold. Therefore, we may impose the condition $x, y, z \leq \frac{1}{2}$. Then the condition $x, y, z \neq 1$ is redundant.

Now, we fix any $z \leq \frac{1}{2}$. Define

$$f(x, y) = \frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2}$$

and $g(x, y) = xy$. We want to minimize $f(x, y)$ given $g(x, y) = \frac{1}{z}$ and $-|z| \leq x, y \leq \frac{1}{2}$. The minimum value exists by the extreme value theorem.

- For interior points inside the set $S = \left\{ (x, y) : -|z| < x, y < \frac{1}{2}, g(x, y) = \frac{1}{z} \right\}$, we can use the Lagrange multiplier rule. We compute

$$\begin{aligned} \nabla f(x, y) &= \left(-\frac{2x}{(x-1)^3}, -\frac{2y}{(y-1)^3} \right), \\ \nabla g(x, y) &= (y, x) \neq (0, 0). \end{aligned}$$

Then we solve

$$\begin{aligned} -\frac{2x}{(x-1)^3} &= \mu y, \\ -\frac{2y}{(y-1)^3} &= \mu x. \end{aligned}$$

This is the same as the system we solve at the beginning, with $\mu = \lambda z$. Therefore, we have $x = y = \frac{1}{\sqrt{z}}$ or $x^2y^2 - 3xy + x + y = 0$. In the latter case, we can further rewrite this as $x + y = \frac{3}{z} - \frac{1}{z^2}$ using $xyz = 1$.

Rather than comparing the images of the points in the two cases, we just need to prove the desired inequality in both cases. If $x = y = \frac{1}{\sqrt{z}}$, then we have

$$\begin{aligned} \frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2} &\geq 1 - \frac{z^2}{(z-1)^2} \\ \Leftrightarrow \frac{2}{(1-\sqrt{z})^2} &\geq \frac{1-2z}{(z-1)^2} \\ \Leftrightarrow 2(1+\sqrt{z})^2 &\geq 1-2z \\ \Leftrightarrow 4z+4\sqrt{z}+1 &\geq 0 \\ \Leftrightarrow (2\sqrt{z}+1)^2 &\geq 0, \end{aligned}$$

which is obvious.

If $x + y = \frac{3}{z} - \frac{1}{z^2}$, then we have

$$\begin{aligned}
& \frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2} \geq 1 - \frac{z^2}{(z-1)^2} \\
\Leftrightarrow & \frac{2x^2y^2 - 2xy(x+y) + x^2 + y^2}{(xy - x - y + 1)^2} \geq \frac{1-2z}{(z-1)^2} \\
\Leftrightarrow & \frac{-2z^3 + 5z^2 - 4z + 1}{(z^2 - 2z + 1)^2} \geq \frac{1-2z}{(z-1)^2} \\
& \text{(by using } xy = \frac{1}{z} \text{ and } x + y = \frac{3}{z} - \frac{1}{z^2}\text{)} \\
\Leftrightarrow & -2z^3 + 5z^2 - 4z + 1 \geq (1-2z)(z-1)^2.
\end{aligned}$$

One may check that the two sides are indeed equal.

- For boundary points where $x = \frac{1}{2}$ (or $y = \frac{1}{2}$), we have proved the inequality before.
- For boundary points where $x = -|z|$ (or $y = -|z|$), we must have $z > 0$ and $x = -z$ since x, y, z cannot be all negative (from $xyz = 1$). This yields $y = \frac{1}{xz} = -\frac{1}{z^2}$. As $y \geq -z$, we have $1 \leq z^3$, i.e. $z \geq 1$. But this contradicts the assumption $z \leq \frac{1}{2}$.

This proves the inequality. Equality holds when $x + y = \frac{3}{z} - \frac{1}{z^2}$ and $xyz = 1$. It is not hard to solve part (b) after noticing this. We shall omit the details. $\square$

Problem 3.8. (CWMO 2007 Problem 3) Suppose the real numbers a, b, c satisfy $a + b + c = 3$. Prove that

$$\frac{1}{5a^2 - 4a + 11} + \frac{1}{5b^2 - 4b + 11} + \frac{1}{5c^2 - 4c + 11} \leq \frac{1}{4}.$$

Analysis. This time we work on the problem in a different way. The expression is of the form $f(a) + f(b) + f(c)$ for some one variable function f . Together with the constraint $a + b + c = 3$, it seems that Jensen's inequality is suitable to prove such an inequality. However, it turns out that f is not always convex or concave. Although we cannot deduce the convexity of f , we can use the graph of its derivative to solve the Lagrange multiplier equation, which is not easy to solve directly.

Solution. Define

$$f(x) = \frac{1}{5x^2 - 4x + 11},$$

$g(a, b, c) = a + b + c$ and $h(a, b, c) = f(a) + f(b) + f(c)$. As usual, we first prove that h attains a minimum value on $\{(a, b, c) : g(a, b, c) = 3\}$. Note that

$$5x^2 - 4x + 11 = 5\left(x - \frac{2}{5}\right)^2 + \frac{51}{5} \geq \frac{51}{5}.$$

This implies $f(x) \leq \frac{5}{51}$, and hence

$$h(a, b, c) \leq f(a) + \frac{5}{51} + \frac{5}{51} = f(a) + \frac{10}{51}.$$

Since $\lim_{a \rightarrow \pm\infty} f(a) = 0$ and $\frac{10}{51} < \frac{1}{4}$, the inequality holds whenever $|a|$ is large. Therefore, it suffices to consider the compact domain $S = \{(a, b, c) : |a|, |b|, |c| \leq M, g(a, b, c) = 3\}$ for some large constant M . By the extreme value theorem, h attains the minimum value on S . Also, this minimum value must be attained at an interior point (with $|a|, |b|, |c| < M$).

We compute $f'(x) = \frac{-10x + 4}{(5x^2 - 4x + 11)^2}$, so that

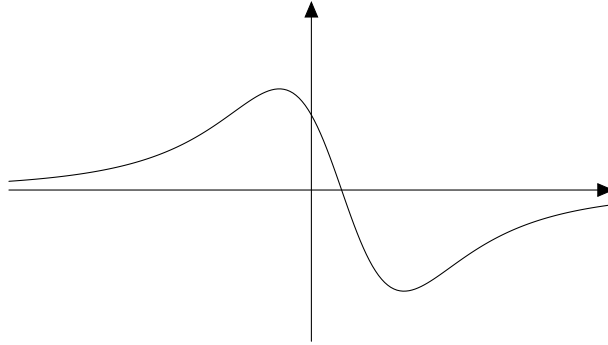
$$\begin{aligned} \nabla h(a, b, c) &= \left(\frac{-10a + 4}{(5a^2 - 4a + 11)^2}, \frac{-10b + 4}{(5b^2 - 4b + 11)^2}, \frac{-10c + 4}{(5c^2 - 4c + 11)^2} \right), \\ \nabla g(a, b, c) &= (1, 1, 1) \neq (0, 0, 0). \end{aligned}$$

The equation $\nabla h(a, b, c) = \lambda \nabla g(a, b, c)$ is simply $f'(a) = f'(b) = f'(c) = \lambda$. However, it is not too easy to solve this system directly.

As all equations are of the form $f'(x) = \lambda$, we can get more information about the solution by studying the function f' . We find that

$$f''(x) = \frac{150x^2 - 120x - 78}{(5x^2 - 4x + 11)^3}.$$

Note that the numerator is a quadratic polynomial with two real roots $\alpha_1 < \alpha_2$, while the denominator is always positive. Therefore, we have $f''(x) > 0$ if $x < \alpha_1$ or $x > \alpha_2$, and $f''(x) < 0$ if $\alpha_1 < x < \alpha_2$. This implies f' is increasing on $(-\infty, \alpha_1)$, decreasing on (α_1, α_2) , and increasing on (α_2, ∞) . Also, it is obvious that f' tends to 0 when x tends to either infinity. The graph $y = f'(x)$ can be roughly sketched as follows.



From the graph, we see that for every λ , there are at most two solutions to the equation $f'(x) = \lambda$. Since we are solving the equation $f'(a) = f'(b) = f'(c) = \lambda$, at least two of a, b, c are equal. WLOG assume $a = b$. Then $c = 3 - 2a$. As the point of maximum of h must satisfy these conditions, it suffices to check that $h(a, b, c) \leq \frac{1}{4}$ whenever $a = b$ and $c = 3 - 2a$. In other words, we need to prove the inequality

$$\frac{2}{5a^2 - 4a + 11} + \frac{1}{5(3 - 2a)^2 - 4(3 - 2a) + 11} \leq \frac{1}{4}.$$

Now,

$$\begin{aligned} & \frac{2}{5a^2 - 4a + 11} + \frac{1}{5(3 - 2a)^2 - 4(3 - 2a) + 11} \leq \frac{1}{4} \\ \Leftrightarrow & \frac{2}{5a^2 - 4a + 11} + \frac{1}{20a^2 - 52a + 44} \leq \frac{1}{4} \\ \Leftrightarrow & 40a^2 - 104a + 88 + 5a^2 - 4a + 11 \leq 25a^4 - 85a^3 + 162a^2 - 187a + 121 \\ \Leftrightarrow & 25a^4 - 85a^3 + 117a^2 - 79a + 22 \geq 0 \\ \Leftrightarrow & (a - 1)(25a^3 - 60a^2 + 57a - 22) \geq 0 \\ \Leftrightarrow & (a - 1)^2(25a^2 - 35a + 22) \geq 0. \end{aligned}$$

(Note that the last factorization is obvious as $a = 1$ is an equality case.) This is true as the discriminant of $25a^2 - 35a + 22$ is $(35)^2 - 4(25)(22) = 1225 - 2200 < 0$. This completes the proof. $\square$

In all the above problems, there is only one equation constraint. Sometimes there may be more than one such constraint, and the Lagrange multiplier rule we stated before cannot be used. Indeed, there is a more general result that can handle this issue.

Theorem 3.2. ([Lagrange multiplier rule](#)(拉格朗日乘子法) — multiple constraints) Let $f, g_1, g_2, \dots, g_k : X \rightarrow \mathbb{R}$ be functions whose partial derivatives are continuous, where $X \subset \mathbb{R}^n$ is open and $k < n$. Define

$$S = \{\mathbf{x} \in X : g_j(\mathbf{x}) = c_j \text{ for } j = 1, 2, \dots, k\}$$

where $c_1, c_2, \dots, c_k$ are constants. If the restriction $f|_S$ of f on S has a local extremum at $\mathbf{a} \in S$ and $\{\nabla g_1(\mathbf{a}), \nabla g_2(\mathbf{a}), \dots, \nabla g_k(\mathbf{a})\}$ is linearly independent, then there exist $\lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{R}$ such that

$$\nabla f(\mathbf{a}) = \lambda_1 \nabla g_1(\mathbf{a}) + \lambda_2 \nabla g_2(\mathbf{a}) + \dots + \lambda_k \nabla g_k(\mathbf{a}).$$

A set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is said to be [linearly independent](#)(線性無關) if the only solution to

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k = \mathbf{0}$$

where $a_1, a_2, \dots, a_k \in \mathbb{R}$ is $a_1 = a_2 = \dots = a_k = 0$. In particular, when $k = 2$, this means each of $\mathbf{v}_1$ and $\mathbf{v}_2$ is not a multiple of the other.

This time we are optimizing the function f under k constraints $g_j(\mathbf{x}) = c_j$. When $k = 1$, this result is reduced to [theorem 3.1](#), so this is a generalization of that. Note that we require $k < n$, which means the number of constraints should be less than the number of variables. Otherwise, the constraints already determine a system with as many equations as the number of variables. Most likely there are only finitely many solutions to such a system.

We provide one MO problem below which can be solved using this generalized version of the Lagrange multiplier rule.

Problem 3.9. (IMO Shortlist 2010 A2) Let the real numbers a, b, c, d satisfy the relations $a + b + c + d = 6$ and $a^2 + b^2 + c^2 + d^2 = 12$. Prove that

$$36 \leq 4(a^3 + b^3 + c^3 + d^3) - (a^4 + b^4 + c^4 + d^4) \leq 48.$$

Analysis. The ideas are exactly the same as before, except that we need to use the Lagrange multiplier rule with two constraints. We first need to prove the existence of global extrema.

Solution. Let

$$\begin{aligned} f(a, b, c, d) &= 4(a^3 + b^3 + c^3 + d^3) - (a^4 + b^4 + c^4 + d^4), \\ g_1(a, b, c, d) &= a + b + c + d, \\ g_2(a, b, c, d) &= a^2 + b^2 + c^2 + d^2. \end{aligned}$$

Clearly, the set $S = \{(a, b, c, d) : g_1(a, b, c, d) = 6, g_2(a, b, c, d) = 12\}$ is compact, and f is continuous. Thus, the global extrema of f on S exist.

Next, we compute

$$\begin{aligned} \nabla f(a, b, c, d) &= (12a^2 - 4a^3, 12b^2 - 4b^3, 12c^2 - 4c^3, 12d^2 - 4d^3), \\ \nabla g_1(a, b, c, d) &= (1, 1, 1, 1), \\ \nabla g_2(a, b, c, d) &= (2a, 2b, 2c, 2d). \end{aligned}$$

The vectors $(1, 1, 1, 1)$ and $(2a, 2b, 2c, 2d)$ are multiples of each other iff $a = b = c = d$. This is rejected since $4a = 6$ and $4a^2 = 12$ cannot hold simultaneously. By the Lagrange multiplier rule, the extremum points must be solutions to the system

$$\begin{aligned} 12a^2 - 4a^3 &= \lambda_1 + 2\lambda_2 a, & (1) \\ 12b^2 - 4b^3 &= \lambda_1 + 2\lambda_2 b, & (2) \\ 12c^2 - 4c^3 &= \lambda_1 + 2\lambda_2 c, \\ 12d^2 - 4d^3 &= \lambda_1 + 2\lambda_2 d. \end{aligned}$$

From (2)–(1), we get

$$(a - b)(4a^2 + 4ab + 4b^2 - 12a - 12b) = 2\lambda_2(b - a).$$

As we have shown that a, b, c, d cannot be all equal, we may assume $a \neq b$. Then this yields

$$a^2 + ab + b^2 - 3a - 3b = -\frac{1}{2}\lambda_2. \quad (3)$$

If $c \neq a$ and $c \neq b$, then similarly we have

$$a^2 + ac + c^2 - 3a - 3c = -\frac{1}{2}\lambda_2. \quad (4)$$

Equating (3) and (4), we get $(b - c)(a + b + c - 3) = 0$. Hence, $a + b + c = 3$. In that case, we have $d = 6 - (a + b + c) = 3$. This in turns implies $a^2 + b^2 + c^2 = 12 - d^2 = 3$. Thus, we have

$$9 = (a + b + c)^2 \leq 3(a^2 + b^2 + c^2) = 9.$$

As equality holds, we have $a = b = c$, contradiction.

It follows that $c = a$ or $c = b$. Similarly, $d = a$ or $d = b$. Up to symmetry, there are two possibilities.

- If $c = a$ and $d = b$, the conditions become $a + b = 3$ and $a^2 + b^2 = 6$. Then $ab = \frac{3}{2}$, and so a, b are roots to $x^2 - 3x + \frac{3}{2} = 0$. This gives $a, b = \frac{3 \pm \sqrt{3}}{2}$, and

$$f(a, b, c, d) = 8\left(\frac{3 + \sqrt{3}}{2}\right)^3 + 8\left(\frac{3 - \sqrt{3}}{2}\right)^3 - 2\left(\frac{3 + \sqrt{3}}{2}\right)^4 - 2\left(\frac{3 - \sqrt{3}}{2}\right)^4 = 45.$$

- If $c = d = a$, the conditions become $3a + b = 6$ and $3a^2 + b^2 = 12$. It is easy to get the two solutions $(a, b, c, d) = (2, 0, 2, 2), (1, 3, 1, 1)$, and we have $f(2, 0, 2, 2) = 48$ and $f(1, 3, 1, 1) = 36$.

By comparing the above values, we see that 36 and 48 are the minimum and maximum values of $f(a, b, c, d)$. This proves the inequality. $\square$

Links

Theorems

- [Extreme value theorem](#)
- [Lagrange multiplier rule \(one constraint\)](#)
- [Lagrange multiplier rule \(multiple constraints\)](#)

Propositions

- [1.1](#): Preimage of an open/closed set under a continuous function
- [1.2](#): Image of a compact set under a continuous function
- [2.1](#): A necessary condition for a point to be a local extremum point

Problems

- [2.1](#): Optimization (bounded and non-compact domain)
- [2.2](#): Optimization (compact domain)
- [2.3](#): Optimization (unbounded domain)
- [2.4](#): Optimization (unbounded set of equality case)
- [2.5](#): APMO 2004 Problem 5
- [3.1](#): Lagrange multiplier (without boundary)
- [3.2](#): Lagrange multiplier (with boundary)
- [3.3](#): CWMO 2004 Problem 8
- [3.4](#): CGMO 2011 Problem 3
- [3.5](#): IMO 2006 Problem 3
- [3.6](#): IMO 2003 Problem 5
- [3.7](#): IMO 2008 Problem 2
- [3.8](#): CWMO 2007 Problem 3
- [3.9](#): IMO Shortlist 2010 A2

Terminologies and Notations

- [accumulation point](#)
- [boundary](#)
- [boundary point](#)
- [bounded set](#)

- closed ball $\overline{B}(\mathbf{a}, r)$
- closed disc
- closed set
- compact set
- component function
- continuous
- continuous function
- discontinuous
- Euclidean space $\mathbb{R}^n$
- extremum
- global maximum
- global minimum
- gradient ∇f
- interior
- interior point
- isolated point
- length $\|\mathbf{x}\|$, $|\mathbf{x}|$
- limit $\lim_{\mathbf{x} \rightarrow \mathbf{a}} \mathbf{f}(\mathbf{x})$
- limit point
- linearly independent
- local maximum
- local minimum
- norm $\|\mathbf{x}\|$, $|\mathbf{x}|$
- open ball $B(\mathbf{a}, r)$
- open disc
- open set
- partial derivative $\frac{\partial f}{\partial x_j}$, f_{x_j}
- vector $\mathbf{v}$
- zero vector $\mathbf{0}$

「真正的人、真正的事，往往不及心中所想的那麼好。」

《倚天屠龍記》